

# Degenerative Lumbosacral Stenosis—Conservative Management\*—cont'd

| Stage                                                                                                                                                                                                                                                                          | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outcome Assessment Measures                                                                                                                                                                                                                                                                                                                                                                                 |
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|                                                                                                                                                                                                                                                                                | <p><b>Modalities</b></p> <ul style="list-style-type: none"> <li>Warm packs across L7-S1 BID-TID or before exercise PRN</li> <li>TENS across L7-S1 15-20 min SID-EOD</li> <li>Gentle massage of lumbar epaxial and limb musculature SID-BID</li> <li>Extracorporeal shock wave treatment</li> </ul> <p><b>Medications</b></p> <ul style="list-style-type: none"> <li>NSAIDs, NMDA antagonists and other analgesics PRN</li> <li>Epidural corticosteroid injection protocol if needed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>Modalities</b></p> <ul style="list-style-type: none"> <li>Warm packs across L7-S1 BID-TID or before exercise PRN</li> <li>Effleurage BID-TID before exercise PRN</li> <li>Cryotherapy to paraspinal region of injury 20 min after exercise PRN</li> </ul> <p><b>Medications</b></p> <ul style="list-style-type: none"> <li>NSAIDs, NMDA antagonists and other analgesics PRN</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Moderate Pain**</b></p> <p><b>Goals:</b></p> <p>Mild-to-minimal pain with activity</p> <p>Controlled navigation on challenging surfaces</p> <p>Prevent/minimize limb muscle atrophy</p> <p>Maintain normal limb ROM</p> <p>Early strengthening of paraspinal muscles</p> | <p><b>Therapeutic Exercises</b></p> <p><i>Introduce gradually as tolerated by the patient</i></p> <ul style="list-style-type: none"> <li>Balance on a bi-directional balance board (all four limbs) or semi-inflated disk (hindlimbs) 5 min SID-BID</li> <li>Aquatic therapy: Underwater Treadmill (UWTM) walking, 5-10 min working up to 15-20 min SID slow, steady pace</li> <li>Flexor reflex exercises for affected limbs (10-15 reps BID-TID)</li> <li>Walking through weave poles, gradually decreasing distance, 5-10 laps through 6-8 poles SID</li> <li>Low Cavaletti rails 5-10 reps SID-BID</li> <li>Sit-to-stand 5 working up to 15-20 reps SID-BID</li> <li>Balance with pelvic limbs on a high-traction surface and antebrachii or forepaws balancing on a physioroll or peanut. Apply gradually increasingly challenging displacement forces in multiple directions to encourage walking of the hindlimbs while balancing. 5 min SID.</li> </ul> <p><i>Select an appropriately sized physioroll to prevent spinal hyperextension with this activity.</i></p> <p><b>Caution:</b> Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</p> | <p><b>Therapeutic Exercises</b></p> <p><i>Introduce gradually as tolerated by the patient</i></p> <ul style="list-style-type: none"> <li>Balance with very gentle displacement forces applied across a mildly unstable surface (eg sofa cushion) 5 min SID-BID</li> <li>Flexor reflex exercises for affected limbs (10-15 reps BID-TID)</li> <li>Walking in circles and figure eights, gradually decreasing size, 5-10 min SID-BID</li> <li>Walking in sand, snow, low water or across a foam mattress 1-3 min 2-3 times per week.</li> <li>Sit-to-stand 5 working up to 15-20 reps SID-BID</li> <li>Moderate leash walks 10-20 min BID, incorporating 5 minutes of mild inclines</li> </ul> <p><b>Caution:</b> Choose only 2-4 exercises per session. Doing all exercises may result in over-exertion and pain early in the program.</p> <p><b>Modalities</b></p> <ul style="list-style-type: none"> <li>Warm packs across L7-S1 BID-TID or before exercise PRN</li> <li>Effleurage BID-TID before exercise PRN</li> <li>Cryotherapy to paraspinal region of injury 20 min after exercise PRN</li> </ul> | <ul style="list-style-type: none"> <li>Subjective pain level at digital pressure L7-S1, dorsal tail elevation, back hyperextension, and rectal palpation</li> <li>Hindlimb proprioception and postural reactions</li> <li>Thigh circumference</li> <li>Ability to navigate obstacles and mild inclines at a walk</li> </ul> <p><i>If consistent pain control is achieved move to mild pain protocol</i></p> |

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| <p>Modalities</p> <ul style="list-style-type: none"> <li>• Warm packs across L7-S1 BID-TID or before exercise PRN</li> <li>• TENS across L7-S1 15-20 min SID-EOD</li> <li>• Extracorporeal shock wave treatment</li> <li>• Therapeutic laser lumbar epaxial muscles SID</li> <li>• Gentle massage of lumbar epaxial and limb musculature SID-BID</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs, NMDA antagonists and other analgesics PRN</li> <li>• Muscle relaxants PRN if severe epaxial/pelvic limb muscle spasms</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs, NMDA antagonists and other analgesics PRN</li> <li>• Epidural corticosteroid injection protocol if needed</li> <li>• Muscle relaxants PRN if severe epaxial/pelvic limb muscle spasms</li> </ul> | <p>Severe Pain**</p> <p><u>Goals:</u></p> <p>Pain and muscle spasm reduction</p> <p>Maintain normal limb ROM</p> <p>Increase affected limb muscle mass</p> <p>Maintenance of balance during turns and on a moderately unstable surface</p> <p>Therapeutic Exercises</p> <p><i>Introduce gradually as tolerated by the patient</i></p> <ul style="list-style-type: none"> <li>• Aquatic therapy: Underwater Treadmill (UWTM) walking, 2-5 min working up to 10-15 min SID slow, steady pace</li> <li>• Flexor reflex exercises for affected limbs (10-15 reps BID-TID)</li> <li>• PROM pelvic limbs 10-15 reps BID-TID</li> <li>• Low Cavaletti Rails 5 reps SID-BID</li> <li>• Sit-to-stand 5 reps SID-BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i></p> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Warm packs across L7-S1 BID-TID or before exercise PRN</li> <li>• TENS across L7-S1 15-20 min SID-EOD</li> <li>• Therapeutic laser lumbar epaxial muscles SID</li> <li>• Gentle massage of lumbar epaxial (if tolerated) and limb musculature SID-BID</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs, NMDA antagonists, and other analgesics PRN</li> <li>• Epidural corticosteroid injection protocol if needed</li> <li>• Muscle relaxants PRN if severe epaxial/pelvic limb muscle spasms</li> </ul> <p>Therapeutic Exercises</p> <p><i>Introduce gradually as tolerated by the patient</i></p> <ul style="list-style-type: none"> <li>• Balance with very gentle displacement forces applied across a mildly unstable surface (eg sofa cushion) 5 min SID-BID</li> <li>• Flexor reflex exercises for affected limbs (10-15 reps BID-TID)</li> <li>• PROM pelvic limbs 10-15 reps BID-TID</li> <li>• Walking in circles and figure eights, gradually decreasing size, 5-10 min SID-BID</li> <li>• Walking in sand or on uneven ground 1-3 min 2-3 times per week.</li> <li>• Sit-to-stand 5 reps BID</li> <li>• Slow leash walks 5-10 min BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Doing all exercises may result in over-exertion and pain early in the program.</i></p> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Warm packs across L7-S1 BID-TID or before exercise PRN</li> <li>• Effleurage BID-TID before exercise PRN</li> <li>• Cryotherapy to paraspinal region of injury 20 min after exercise PRN</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs, NMDA antagonists, and other analgesics PRN</li> <li>• Muscle relaxants PRN if severe epaxial/pelvic limb muscle spasms</li> </ul> <p>• Response to activity and subjective pain level when active and at rest</p> <p>• Length of time with unsupported standing</p> <p>• Length of time walking in aquatic therapy and on dry land</p> <p>• Frequency of steps in which appropriate proprioception is used</p> <p>• Height of objects navigated cleanly</p> <p>• Thigh circumference/muscle mass</p> <p><i>If consistent pain control is achieved move to moderate pain protocol</i></p> |
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This protocol is for use by licensed personnel who have been trained in canine rehabilitation.

\*This protocol is designed for patients for whom conservative management is appropriate. Patients that have sacral subluxation at L7-S1 or an extruded intervertebral disk causing significant cauda equina compression that are good surgical candidates would likely receive greatest benefit from surgical intervention.

\*\*Patients with moderate to severe pain, paraparesis, or ataxia should have appropriate diagnostic imaging and surgery if indicated.

## Elbow Dysplasia\*\*—Post-arthroscopy

| Phase                                                                                                                                                                     | Expected Timeframe*                      | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Outcome Assessment Measures                                                                                                                                                                                                   | Criteria for Movement to Next Phase                                                                                                                                                                                                                                                           |
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| <b>Step 1</b><br>Postoperative to Early Toe-Touching at a Walk<br><u>Goals:</u><br>Pain control<br>Swelling and edema control<br>Maintain/improve elbow postoperative ROM | Immediate to 48-72 hours postoperatively | <b>Therapeutic Exercises</b> <ul style="list-style-type: none"> <li>Slow leash walking only to go outside, no more than five minutes at a time, with a harness</li> <li>Gentle PROM affected thoracic limb (with emphasis on elbow flexion/extension), 8-10 repetitions BID-TID</li> </ul> <b>Modalities</b> <ul style="list-style-type: none"> <li>Gentle massage (effleurage, shoulder muscles)</li> <li>Transcutaneous electrical stimulation across the elbow for pain relief (15-20 min SID-TID)</li> <li>Cryotherapy to affected elbow (15-20 min BID-TID)</li> <li>Therapeutic Laser SID</li> </ul> <b>Medications</b> <ul style="list-style-type: none"> <li>NSAIDs/other analgesics</li> <li>Oral chondroprotective agents</li> <li>Injectable polysulfated glycosaminoglycans</li> </ul> | <b>Therapeutic Exercises</b> <ul style="list-style-type: none"> <li>Slow leash walking only to go outside, no more than five minutes at a time, with a harness</li> <li>Gentle PROM affected thoracic limb (with emphasis on elbow flexion/extension), 8-10 repetitions BID-TID</li> </ul> <b>Modalities</b> <ul style="list-style-type: none"> <li>Gentle massage (effleurage)</li> <li>Cryotherapy to affected elbow (15-20 min BID-TID)</li> </ul> <b>Medications</b> <ul style="list-style-type: none"> <li>NSAIDs/other analgesics</li> <li>Oral chondroprotective agents</li> <li>Injectable polysulfated glycosaminoglycans</li> </ul> | <ul style="list-style-type: none"> <li>Postoperative elbow PROM (other joints if applicable)—goniometry</li> <li>Response to activity and subjective pain level</li> <li>Gait evaluation at a walk</li> <li>Weight</li> </ul> | <ul style="list-style-type: none"> <li>Consistent toe-touching on operated limb</li> <li>Adequate resting analgesia</li> <li>Decreased perioperative swelling and no incisional drainage</li> <li>Comfortable elbow ROM</li> <li>80-100 degrees flexion, 135-150 degrees extension</li> </ul> |
| <b>Step 2</b><br>Early Weight-Bearing, Moderate Elbow Flexion<br><u>Goals:</u><br>Continue pain control<br>Improve elbow ROM                                              | 48-72 hours to two weeks postoperatively | <b>Therapeutic Exercises</b> <ul style="list-style-type: none"> <li>Land Treadmill: Begin with slight incline or level surface, 5-10 minutes at a slow, steady pace BID with a harness</li> <li>PROM within comfortable range affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions BID-TID</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Therapeutic Exercises</b> <ul style="list-style-type: none"> <li>Slow, steady leash walking on flat ground 5 min working up to 15 minutes at a time with a harness</li> <li>PROM within comfortable range affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions BID-TID</li> </ul>                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Response to light activity and subjective pain level</li> <li>Gait evaluation at walk</li> <li>Goniometry—Elbow ROM and other joints if applicable</li> </ul>                          | <ul style="list-style-type: none"> <li>Consistent partial weight-bearing with each stride on operated limb</li> <li>Minimal pain with light activities</li> <li>Incisional healing without complications</li> </ul>                                                                           |

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| <p>Increase toe-touching weight-bearing Prevent/minimize thoracic limb muscle atrophy</p> | <ul style="list-style-type: none"> <li>• Balance/weight-shifting exercises on a bi-directional board or soft foam pad 5-10 min BID-TID</li> <li>• “Shaking paw” exercises (begin training if applicable) 8-10 reps BID-TID</li> <li>• Flexor withdrawal stimulation 8-10 reps BID affected thoracic limb</li> <li>Modalities             <ul style="list-style-type: none"> <li>• Transcutaneous electrical stimulation across the elbow for pain relief (15-20 min PRN)</li> <li>• NMES muscles affected thoracic limb 15 min SID/PRN for atrophy prevention</li> <li>• Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)</li> <li>• Therapeutic Laser SID</li> </ul> </li> <li>Medications             <ul style="list-style-type: none"> <li>• NSAIDs/other analgesics</li> <li>• Oral chondroprotective agents</li> <li>• Injectable polysulfated glycosaminoglycans</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>• Balance/weight-shifting exercises for thoracic limbs on a semi-firm surface 5-10 min BID-TID</li> <li>• “Shaking paw” exercises (begin training if applicable) 8-10 reps BID-TID</li> <li>• Flexor withdrawal stimulation 8-10 reps BID affected thoracic limb</li> <li>Modalities             <ul style="list-style-type: none"> <li>• Gentle massage (effleurage, scapular muscles)</li> <li>• Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)</li> </ul> </li> <li>Medications             <ul style="list-style-type: none"> <li>• NSAIDs/other analgesics</li> <li>• Oral chondroprotective agents</li> <li>• Injectable polysulfated glycosaminoglycans</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>• Subjective thoracic limb muscle mass</li> <li>• Weight</li> <li>• Elbow ROM 60-80 degrees flexion, 145-155 degrees extension</li> </ul>                                                                                     |
| <p><b>Step 3</b><br/>Weight-Bearing at a Walk, Near-Normal Elbow Flexion</p>              | <p>Two to four weeks postoperatively</p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Land Treadmill: Begin with slight incline or level surface, 5-10 minutes at a slow, steady pace BID with a harness</li> <li>• PROM within comfortable range affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions BID-TID</li> <li>• Aquatic Therapy: Underwater Treadmill (UWTM) slow walking 5-10 min SID-BID</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Weight loss program can be initiated at this phase if warranted (minimizes loss of lean tissue mass).</p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, steady leash walking on flat ground 5 min working up to 15 minutes at a time with a harness</li> <li>• PROM within comfortable range affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions BID-TID</li> </ul>                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Response to moderate activity and subjective pain level</li> <li>• Gait evaluation at walk</li> <li>• No evidence of pain with moderate activities</li> <li>• Consistent weight-bearing on operated limb at walk</li> </ul> |

Continued

## Elbow Dysplasia\*\*—Post-arthroscopy—cont'd

| Phase                                                                        | Expected<br>Timeframe* | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                    | Home Program                                                                                                                                                                    | Outcome<br>Assessment<br>Measures                                                                     | Criteria for<br>Movement to Next<br>Phase                                                                     |
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| Goals:<br>Minimize pain during exercises                                     |                        | <ul style="list-style-type: none"> <li>Swimming 5-10 minutes 3-4 times per week with breaks (<i>Note: avoid this activity until 4-6 weeks postoperatively if excessive force observed with forelimb movements</i>)</li> </ul>                                                                                                    | <ul style="list-style-type: none"> <li>Balance/weight-shifting exercises for thoracic limbs on a semi-unstable surface 5-10 min BID-TID</li> </ul>                              | <ul style="list-style-type: none"> <li>Goniometry—Elbow ROM and other joints if applicable</li> </ul> | <ul style="list-style-type: none"> <li>Elbow PROM 40-60 degrees flexion, &gt;150 degrees extension</li> </ul> |
| Achieve near-normal elbow ROM (unless significant osteoarthritis is present) |                        | <ul style="list-style-type: none"> <li>Balance/weight-shifting exercises on a 360-degree balance board 5-10 min BID-TID</li> </ul>                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>“Shaking paw” exercises 10-15 reps BID-TID. Add light weights to proximal antebrachium as tolerated for muscle strengthening.</li> </ul> | <ul style="list-style-type: none"> <li>Subjective thoracic limb muscle mass</li> </ul>                |                                                                                                               |
| Improve weight-bearing at a walk                                             |                        | <ul style="list-style-type: none"> <li>“Shaking paw” exercises 10-15 reps BID-TID. Add light weights to proximal antebrachium as tolerated for muscle strengthening.</li> </ul>                                                                                                                                                  | <ul style="list-style-type: none"> <li>Flexor withdrawal stimulation 8-10 reps BID affected thoracic limb</li> </ul>                                                            | <ul style="list-style-type: none"> <li>Weight</li> </ul>                                              |                                                                                                               |
| Improve operated limb muscle mass/strength                                   |                        | <ul style="list-style-type: none"> <li>Flexor withdrawal stimulation 8-10 reps BID affected thoracic limb</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> | <ul style="list-style-type: none"> <li>Heat therapy at elbow prior to PROM exercises 10 min (if no signs of acute inflammation present)</li> </ul>                              |                                                                                                       |                                                                                                               |
|                                                                              |                        | <p><i>per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p>                                                                                                                                                         | <ul style="list-style-type: none"> <li>Gentle massage (effleurage, scapular muscles)</li> </ul>                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        | <p><i>exercises may result in lameness and pain early in the program</i></p>                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)</li> </ul>                                                          |                                                                                                       |                                                                                                               |
|                                                                              |                        | Modalities                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        | <ul style="list-style-type: none"> <li>Heat therapy at elbow prior to PROM exercises 10 min (if no signs of acute inflammation present)</li> </ul>                                                                                                                                                                               |                                                                                                                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        | <ul style="list-style-type: none"> <li>Transcutaneous electrical stimulation across the elbow for pain relief (15-20 min PRN)</li> </ul>                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        | <ul style="list-style-type: none"> <li>NMES thoracic limb muscles affected thoracic limb 15 min PRN for atrophy prevention</li> </ul>                                                                                                                                                                                            |                                                                                                                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        | <ul style="list-style-type: none"> <li>Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)</li> </ul>                                                                                                                                                                                                           |                                                                                                                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        | <ul style="list-style-type: none"> <li>Therapeutic Laser PRN</li> </ul>                                                                                                                                                                                                                                                          |                                                                                                                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        |                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>NSAIDs/other analgesics</li> </ul>                                                                                                       |                                                                                                       |                                                                                                               |
|                                                                              |                        |                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Oral chondroprotective agents</li> </ul>                                                                                                 |                                                                                                       |                                                                                                               |
|                                                                              |                        |                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Injectable polysulfated glycosaminoglycans</li> </ul>                                                                                    |                                                                                                       |                                                                                                               |

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| Medications                                               |                                                                                                                                                                        |                                                                                                                                                                                                      |                                                                                                                                    |                                                                                                 |                                                        |
|                                                           | <ul style="list-style-type: none"><li>• NSAIDs/other analgesics</li><li>• Oral chondroprotective agents</li><li>• Injectable polysulfated glycosaminoglycans</li></ul> |                                                                                                                                                                                                      |                                                                                                                                    |                                                                                                 |                                                        |
| <u>Step 4</u>                                             | Four to eight weeks                                                                                                                                                    | Therapeutic Exercises                                                                                                                                                                                | Continue weight loss program if needed                                                                                             | Response to moderate/heavy activity and subjective pain level                                   | Minimal to no lameness at a walk                       |
| Consistent Weight-Bearing at a Walk, Normal Elbow Flexion | postoperatively                                                                                                                                                        | • Land Treadmill: Level surface working up to slight decline, walking 10-15 minutes SID-BID with a harness                                                                                           | Therapeutic Exercises                                                                                                              | • Slow, steady leash walking on flat ground 5 min working up to 15 min at a time with a harness | • Partial weight-bearing at a trot                     |
| <u>Goals:</u>                                             |                                                                                                                                                                        | • PROM affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions PRN if ROM is not within normal limits                                                                   | • PROM within comfortable range affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions BID-TID       | • Goniometry—Elbow ROM and other joints if applicable                                           | • No evidence of pain with moderate-intense activities |
| Normal limb use at a walk                                 |                                                                                                                                                                        | • Aquatic Therapy: Underwater Treadmill (UWTM) walking 10-15 min SID-BID, gradually introducing light weights/flotation device on distal affected thoracic limb in the water                         | • Balance/weight-shifting exercises for thoracic limbs on a semi-unstable surface 5-10 min BID-TID                                 | • Gait evaluation at walk and trot                                                              | • Normal elbow range of motion                         |
| Early weight-bearing at a trot                            |                                                                                                                                                                        | • Swimming 10-15 minutes 3-4 times per week with breaks, gradually introducing light weights/flotation device on distal affected thoracic limb in the water                                          | • “Shaking paw” exercises 10-15 reps BID-TID. Light weights to distal antebrachium as tolerated for muscle strengthening           | • Subjective thoracic limb muscle mass                                                          |                                                        |
| Improve thoracic limb muscle mass and strength            |                                                                                                                                                                        | • Balance/weight-shifting exercises on a 360-degree balance board 5-10 min BID-TID                                                                                                                   | • Wheelbarrowing 5-8 reps of 10-15 seconds BID. <i>Begin at 6-8 weeks postoperatively, as feasible depending on patient weight</i> | • Weight                                                                                        |                                                        |
| Improve stamina and endurance                             |                                                                                                                                                                        | • “Shaking paw” exercises 10-15 reps BID-TID. Light weights to distal antebrachium as tolerated for muscle strengthening                                                                             | <i>Caution: Choose only 2-4 exercises per session. Doing all exercises may result in lameness and pain early in the program</i>    |                                                                                                 |                                                        |
|                                                           |                                                                                                                                                                        | • Wheelbarrowing 5-8 reps of 10-15 seconds BID. <i>Begin at 6-8 weeks postoperatively, as feasible depending on patient weight</i>                                                                   |                                                                                                                                    |                                                                                                 |                                                        |
|                                                           |                                                                                                                                                                        | • Low Cavaletti Rails 5-10 reps BID                                                                                                                                                                  | Modalities                                                                                                                         |                                                                                                 |                                                        |
|                                                           |                                                                                                                                                                        | <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i> | • Heat therapy at elbow prior to PROM exercises 10 min                                                                             |                                                                                                 |                                                        |
|                                                           |                                                                                                                                                                        |                                                                                                                                                                                                      | • Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)                                                             |                                                                                                 |                                                        |

Continued



## Elbow Dysplasia\*\*—Post-arthroscopy—cont'd

| Phase                                                                                                                                                                                                                                                    | Expected<br>Timeframe*                                  | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outcome<br>Assessment<br>Measures                                                                                                                                                                                                                                            | Criteria for<br>Movement to Next<br>Phase                                                                                                                                                            |
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|                                                                                                                                                                                                                                                          |                                                         | <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy at elbow prior to PROM exercises 10 min</li> <li>• Transcutaneous electrical stimulation across the elbow for pain relief (15-20 min PRN)</li> <li>• Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)</li> <li>• Therapeutic Laser PRN</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs PRN</li> <li>• Oral chondroprotective agents</li> <li>• Injectable polysulfated glycosaminoglycans</li> </ul>                                                                                                                                               | <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs PRN</li> <li>• Oral chondroprotective agents</li> <li>• Injectable polysulfated glycosaminoglycans</li> </ul>                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                      |
| <p><b>Step 5</b></p> <p>Consistent Weight-Bearing at a Trot</p> <p>Goals:<br/>Trot working up to faster gaits with minimal to no lameness</p> <p>Continue improving muscle mass, strength, and power</p> <p>Continue improving stamina and endurance</p> | <p>Eight to twelve weeks postoperatively and beyond</p> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Land Treadmill: Trotting at a level surface working up to a decline, 5-15 min SID-BID with a harness</li> <li>• PROM affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions PRN if ROM is not within normal limits</li> <li>• Aquatic Therapy: Underwater Treadmill (UWTM) walking 20-30 min SID-BID at moderate/variable speeds, light weights/flotation device on distal affected thoracic limb in the water</li> <li>• Swimming 15-20 minutes 3-4 times per week with breaks, light weights/flotation device on distal affected thoracic limb in the water</li> </ul> | <p>Continue weight loss program if needed</p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, steady leash walking on flat ground 5 min working up to 15 minutes at a time with a harness</li> <li>• PROM within comfortable range affected thoracic limb (with emphasis on elbow flexion/extension), 10-15 repetitions BID-TID</li> <li>• Balance/weight-shifting exercises for thoracic limbs on a semi-unstable surface 5-10 min BID-TID</li> </ul> | <ul style="list-style-type: none"> <li>• Response to increasing activity and subjective pain level</li> <li>• Gait evaluation at walk and trot</li> <li>• Subjective thoracic limb muscle mass evaluation every 2 weeks until symmetry achieved</li> <li>• Weight</li> </ul> | <p>Progressions may be continued as long as lameness or pain with activity is not observed.</p> <p>If this occurs, decrease activity by 50% for 1 week and then increase again by 10-20% weekly.</p> |

Return to as  
normal activity  
as possible

- Balance/weight-shifting exercises on a 360-degree balance board 10 min BID-TID
- “Shaking paw” exercises 10-15 reps BID-TID. Weights to distal antebrachium as tolerated for muscle strengthening
- Wheelbarrowing 5-8 reps of 10-15 seconds BID. *Begin at 6-8 weeks postoperatively, as feasible depending on patient weight*
- Climbing up and down stairs (slow, controlled on harness and leash) 1-2 flights working up to 5-6 flights BID
- Cavaletti Rails 10-15 reps BID
- “Shaking paw” exercises 10-15 reps BID-TID. Weights to distal antebrachium as tolerated for muscle strengthening
- Wheelbarrowing 5-8 reps of 10-15 seconds BID. *Begin at 6-8 weeks postoperatively, as feasible depending on patient weight*
- Climbing up and down stairs (slow, controlled on harness and leash) 1-2 flights working up to 5-6 flights BID
- Controlled ball play 5 min working up to 15 min

#### SID-BID

*Caution: Choose only 2-4 exercises per session. Doing all exercises may result in lameness and pain early in the program*

#### Modalities

- Heat therapy at elbow prior to PROM exercises 10 min PRN
- Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)

#### Medications

- NSAIDs PRN
- Oral chondroprotective agents
- Injectable polysulfated glycosaminoglycans

- Balance/weight-shifting exercises on a 360-degree balance board 10 min BID-TID
- “Shaking paw” exercises 10-15 reps BID-TID. Weights to distal antebrachium as tolerated for muscle strengthening
- Wheelbarrowing 5-8 reps of 10-15 seconds BID. *Begin at 6-8 weeks postoperatively, as feasible depending on patient weight*
- Climbing up and down stairs (slow, controlled on harness and leash) 1-2 flights working up to 5-6 flights BID
- Cavaletti Rails 10-15 reps BID

*Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program*

#### Modalities

- Heat therapy at elbow prior to PROM exercises 10 min PRN
- Cryotherapy to affected elbow (15-20 min BID-TID PRN after exercise)

#### Extracorporeal shock wave

#### Medications

- NSAIDs PRN
- Oral chondroprotective agents
- Injectable polysulfated glycosaminoglycans

This protocol is for use only by licensed personnel who have been trained in canine rehabilitation.

\*The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

\*\*This condition is comprised of one or more of the following: osteochondritis dissecans of the medial humeral condyle, fragmented medial coronoid process, ununited anconeal process.



Fibrocartilaginous Embolic Myelopathy<sup>4,45</sup>

| Phase                                                                                                                                                                                       | Expected Timeframe*                          | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Outcome Assessment Measures                                                                                                                                                                                          | Criteria for Movement to Next Phase                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial Weight Support<br><u>Goals:</u><br>Appreciable limb tone<br>Weight-bearing in standing with minimal support<br>Minimize limb muscle atrophy<br>Independent urination and defecation | Immediate to three to seven days post-injury | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing with support as needed on a stable, high-traction surface (<i>beginning 24 hours post-injury</i>)</li> <li>PROM with bicycling movements all joints of pelvic (+/- thoracic) limbs (10-15 reps BID-TID)</li> <li>Aquatic therapy: Underwater Treadmill (UWTM) standing, 3-5 min SID-BID, with personnel assistance in the water as needed</li> <li>Flexor reflex exercises for affected limbs (10-15 reps BID-TID)</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>Cryotherapy to paraspinal region of injury 20 min TID if pain assessed</li> <li>TENS to paraspinal region of injury 20 min SID-BID if pain assessed</li> <li>NMES to pelvic limb (+/- thoracic limb) musculature for contraction 15 min SID</li> <li>Gentle massage of limb musculature +/- tapotement BID as tolerated</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>Muscle relaxants BID-TID PRN</li> </ul> <p>Other</p> <ul style="list-style-type: none"> <li>Bladder expression TID-QID PRN</li> <li>Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers</li> </ul> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing with support as needed on a stable, high-traction surface (<i>beginning 24 hours post-injury</i>)</li> <li>PROM with bicycling movements all joints of pelvic (+/- thoracic) limbs (10-15 reps BID-TID)</li> <li>Flexor reflex exercises for affected limbs (10-15 reps BID-TID)</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>Cryotherapy to paraspinal region of injury 20 min TID if needed</li> <li>Effleurage BID-TID Medications</li> <li>Muscle relaxants BID-TID PRN</li> </ul> <p>Other</p> <ul style="list-style-type: none"> <li>Bladder expression TID-QID PRN</li> <li>Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers</li> </ul> | <ul style="list-style-type: none"> <li>Subjective comfort level</li> <li>Pelvic (and thoracic, if affected) limb muscle tone</li> <li>Thigh circumference</li> <li>Degree of support required in standing</li> </ul> | <ul style="list-style-type: none"> <li>Standing in weight-bearing with minimal support</li> <li>Initial voluntary stepping movement when attempting to walk</li> </ul> |

| Weight Support<br>to Initial<br>Motor<br>Function<br><u>Goals:</u><br>Pain-free activity<br>Full weight-<br>bearing in<br>standing<br>Minimize limb<br>muscle<br>atrophy<br>Movement of all<br>four limbs at a<br>slow,<br>supported<br>walk<br>Self-rotation | One to two<br>weeks<br>post-injury | Therapeutic Exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Therapeutic Exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Response to<br>activity and<br>subjective pain/<br>effort level<br>Length of time<br>with<br>unsupported<br>standing<br>Number of<br>steps in which<br>gait patterning<br>is used<br>Thigh<br>circumference/<br>muscle mass | Limb<br>voluntary<br>motor activity<br>Unsupported<br>standing |
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|                                                                                                                                                                                                                                                               |                                    | <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting/rhythmic stabilization, gradually decreasing support (eg physioball beneath abdomen, standing in water)</li> <li>PROM with bicycling movements all joints of pelvic (+/- thoracic) limbs (10-15 reps BID-TID)</li> <li>Aquatic therapy: Supported swimming 2-5 minutes working up to 10 minutes with supported standing during breaks. Personnel accompaniment in water</li> <li>Aquatic therapy: Underwater Treadmill (UWTM) assisted walking, 5-15 min SID-BID</li> <li>Turning exercises during standing with positive reinforcement held at various locations (concentrate efforts on the weaker side), encouraging stepping movements towards a reward</li> <li>Slow leash walking with sling support 5 minutes TID, encouraging stepping</li> </ul> | <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting/rhythmic stabilization, gradually decreasing support (eg physioball beneath abdomen, standing on non-slick surface in the bathtub)</li> <li>PROM with bicycling movements all joints of pelvic (+/- thoracic) limbs (10-15 reps BID-TID)</li> <li>Slow leash walking with sling support 5 minutes TID, encouraging stepping</li> <li>Flexor reflex exercises for affected limbs (10-15 reps BID-TID)</li> </ul> |                                                                                                                                                                                                                             |                                                                |
|                                                                                                                                                                                                                                                               |                                    | <p>Modalities</p> <ul style="list-style-type: none"> <li>Gentle massage of affected limb musculature BID-TID</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>Muscle relaxants BID-TID PRN</li> </ul> <p>Other</p> <ul style="list-style-type: none"> <li>Bladder expression TID-QID PRN</li> <li>Use of soft bedding with rotation every 4-6 hours if not self-rotating</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Modalities</p> <ul style="list-style-type: none"> <li>Gentle massage of hindlimb musculature BID-TID</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>Muscle relaxants BID-TID PRN</li> </ul> <p>Other</p> <ul style="list-style-type: none"> <li>Bladder expression TID-QID PRN</li> <li>Use of soft bedding with rotation every 4-6 hours if not self-rotating</li> </ul>                                                                                           |                                                                                                                                                                                                                             |                                                                |

Continued

## Fibrocartilaginous Embolic Myelopathy\*\*—cont'd

| Phase                                                                                                                                                                                                  | Expected<br>Timeframe*              | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Outcome<br>Assessment<br>Measures                                                                                                                                                                                                                                                                                                                                                   | Criteria for<br>Movement to<br>Next Phase                                                                                                            |
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| Initial Motor<br>Function to<br>Good Motor<br>Function with<br>Proprioceptive<br>Deficits                                                                                                              | Two to four<br>weeks<br>post-injury | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting on a bi-directional balance board</li> <li>PROM with bicycling movements all joints of pelvic (+/- thoracic) limbs (10-15 reps PRN based on ROM)</li> <li>Aquatic therapy: Swimming 10-15 minutes with supported standing during breaks</li> <li>Aquatic therapy: Underwater Treadmill (UWTM) walking, 10-15 minutes SID. Use flotation devices/resistance bands to assist with proprioceptive placing if needed.</li> <li>Gait patterning on a ground treadmill, 5-10 minutes SID with resistance band use PRN</li> <li>Turning exercises during standing with positive reinforcement held at various locations (concentrate efforts on the weaker side)</li> <li>Turning exercises during walking: circles and figure eights 5-10 reps SID-BID</li> <li>Low Cavaletti rails 5-10 reps SID-BID</li> <li>Sit-to-stand exercises 5-10 reps BID with hindquarters on an elevated surface or with sling support</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i></p> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting on a moderately unstable surface</li> <li>PROM with bicycling movements all joints of pelvic (+/- thoracic) limbs (10-15 reps BID-TID PRN)</li> <li>Slow leash walking with sling available 5-10 minutes BID-TID</li> <li>Turning exercises during standing with positive reinforcement held at various locations (concentrate efforts on the weaker side)</li> <li>Turning exercises during walking: circles and figure eights 5-10 reps SID-BID</li> <li>Walking across an unstable surface such as a mattress 5-10 reps BID</li> <li>Walking over 5-8 evenly spaced low obstacles 5-10 reps BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i></p> | <ul style="list-style-type: none"> <li>Response to activity and subjective pain level</li> <li>Length of time with unsupported standing</li> <li>Length of time walking in aquatic therapy and on dry land</li> <li>Frequency of steps in which appropriate proprioception is used</li> <li>Height of Cavaletti rails navigated</li> <li>Thigh circumference/muscle mass</li> </ul> | <ul style="list-style-type: none"> <li>Consistent strong limb movements at a walk</li> <li>Minimal over-exertion with moderate activities</li> </ul> |
| Goals:<br>Improve proprioception during walking activity<br>Maintain normal limb ROM<br>Increase affected limb muscle mass<br>Maintenance of balance during turns and on a moderately unstable surface |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                      |
| Modalities                                                                                                                                                                                             |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>NMES to hindlimb musculature for contraction 15 min EOD-SID PRN</li> </ul>                                                                                      |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                      |

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| Good motor function with proprioceptive deficits to near normal gait<br><u>Goals:</u><br>Consistent proprioception with moderate- to high-intensity activity<br>Affected limb muscle mass approaching normal<br>Maintain balance during turns and negotiating obstacles while walking<br>Consistent gait-patterning in early trotting | Four to six weeks post-injury | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting on a 360-degree balance board</li> <li>Aquatic therapy: Swimming 20-30 minutes with supported standing during breaks</li> <li>Aquatic therapy: Underwater Treadmill (UWTM) walking, 15-20 minutes</li> <li>Gait patterning on a ground treadmill, 5-10 minutes with resistance band use PRN. Introduce trotting for 1-2 minutes.</li> <li>Trotting 2-3 minutes over even ground BID</li> <li>Turning exercises during walking: weave poles gradually decreasing in distance 5-10 reps SID-BID</li> <li>Cavaletti rails 10 reps SID-BID</li> <li>Stair Climbing 1 working up to 5 flights SID-BID</li> <li>Sit-to-stand exercises, 5-10 reps SID-BID<br/><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></li> </ul> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting on an unstable surface</li> <li>Turning exercises during walking: weave poles gradually decreasing in distance 5-10 reps SID-BID</li> <li>Stair Climbing 1 working up to 5 flights SID-BID</li> <li>Sit-to-stand exercises, rising from an elevated surface 5-10 reps SID-BID</li> <li>Leash walking 10-15 min BID-TID with 5 minutes inclines/declines</li> <li>Trotting 2-3 minutes over even ground BID</li> <li>Walking across an unstable surface such as a mattress 10-15 reps BID</li> <li>Walking over 5-8 evenly spaced low obstacles 10-15 reps BID<br/><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></li> </ul> | <ul style="list-style-type: none"> <li>Response to activity and subjective pain level</li> <li>Length of time with unsupported standing</li> <li>Walking endurance</li> <li>Tightness of turns in which balance is maintained</li> <li>Frequency of steps in which appropriate proprioception is used</li> <li>Height of Cavaletti rails navigated</li> <li>Thigh circumference/muscle mass</li> </ul> | <ul style="list-style-type: none"> <li>Consistent pelvic limb proprioception at a walk</li> <li>Minimal to no over-exertion or pain with extensive activities</li> <li>Normal gait patterning at a trot</li> </ul> |
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## Fibrocartilaginous Embolic Myelopathy\*\*\*—cont'd

| Phase                                                                                                                                                                                                                                   | Expected Timeframe*           | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outcome Assessment Measures                                                                                                                                                                                                                                                                                        | Criteria for Movement to Next Phase                                                                                                                                            |
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| Near Normal Gait to Normal Gait<br><u>Goals:</u><br>Elimination of proprioceptive deficits during high-intensity activity<br>Normal affected limb muscle mass<br>Maintain balance during turns and negotiating obstacles while trotting | Six or more weeks post-injury | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting on a 360-degree balance board</li> <li>Aquatic therapy: Swimming 20-30 minutes</li> <li>Aquatic therapy: Underwater Treadmill (UWTM) walking, 20-30 minutes at variable speeds</li> <li>Gait patterning on a ground treadmill, 5-10 minutes at a trot</li> <li>Turning exercises with weave poles 10-20 reps SID-BID at a trot</li> <li>Cavaletti rails 10-15 reps SID-BID at a trot</li> <li>Stair Climbing 5-10 repetitions SID-BID</li> <li>Sit-to-stand exercises 10-20 reps</li> <li>Controlled ball play</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting on an unstable surface</li> <li>Turning exercises with weave poles 10-20 reps SID-BID at a trot</li> <li>Stair Climbing 5-10 repetitions SID-BID</li> <li>Sit-to-stand exercises 10-20 reps SID-BID</li> <li>Controlled ball play</li> <li>Leash walking 20-30 minutes</li> <li>BID with 10 minutes variable inclines and 5-10 minutes working up to longer periods trotting</li> <li>Trotting across an unstable surface such as a mattress 10-15 reps BID</li> <li>Trotting over 5-8 evenly spaced low obstacles 10-15 reps BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> | <ul style="list-style-type: none"> <li>Response to activity and subjective pain level</li> <li>Walking and trotting endurance</li> <li>Tightness of turns in which balance is maintained</li> <li>Height and distance of Cavaletti rails navigated</li> <li>Pelvic limb thigh circumference/muscle mass</li> </ul> | <ul style="list-style-type: none"> <li>Minimal to no over-exertion with extensive activities</li> <li>Normal gait and proprioception at a trot and at faster speeds</li> </ul> |

This protocol is for use by licensed personnel who have been trained in canine rehabilitation.

\*The therapist must keep in mind that these timelines are intended as guidelines. Rate of recovery will vary widely among patients for reasons that include initial status of nociception and degree of spinal cord injury. Approximately 50% of patients show improvement in the first few weeks.

\*\*Fibrocartilaginous embolism is often difficult to differentiate from acute intervertebral disk extrusion. Appropriate diagnostics should be performed to differentiate the conditions to determine the need for surgical decompression. Caution should be taken and progression of rehabilitation should be slower in patients for which the latter condition may be suspected.

## Femoral Head and Neck Osteotomy

| Phase                                     | Expected<br>Timeframe*                       | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Outcome<br>Assessment<br>Measures                                                                                                                                                                                                                                                                                      | Criteria for<br>Movement to<br>Next Phase                                                                                                                                                |
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| Non-Weight-<br>Bearing to<br>Toe-Touching | Immediate to<br>48-72 hours<br>postoperative | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, gentle hip pseudoarthrosis PROM for operated limb focusing on extension (10 reps TID-QID beginning immediately postoperative while recovering from anesthesia)</li> <li>• Slow leash walking with sling support available, only to go outside (up to 5 minutes, TID-QID)</li> <li>• Balance exercises on a soft foam pad or bi-directional balance board for weight-bearing</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs</li> <li>• Opioids; injectable/CRI</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Gentle massage around the pseudoarthrosis, thigh, and lumbosacral regions</li> <li>• Transcutaneous electrical stimulation for pain relief (15-20 min SID-TID)</li> <li>• Cryotherapy (15-20 min TID after activities)—<i>first session immediately postoperative in combination with slow PROM while recovering from anesthesia</i></li> <li>• Therapeutic laser therapy SID</li> </ul> | <p>Inpatient status preferred during this phase</p> <p><u>If Home:</u></p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, gentle hip pseudoarthrosis PROM for operated limb focusing on extension (10 reps TID-QID beginning immediately postoperative)</li> <li>• Slow leash walking with sling support available, only to go outside (up to 5 minutes, TID-QID)</li> <li>• Balance exercises on a semi-firm surface for weight-bearing</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs</li> <li>• Opioids</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Gentle massage around the pseudoarthrosis, thigh, and lumbosacral regions</li> <li>• Cryotherapy (15-20 min TID after activities)—<i>first session immediately postoperative</i></li> </ul> | <ul style="list-style-type: none"> <li>• Postoperative bilateral “hip” PROM and other joints as applicable via goniometry</li> <li>• Postoperative bilateral thigh circumference</li> <li>• Response to activity and subjective pain level</li> <li>• Lameness score at a stance and walk</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Early toe-touching</li> <li>• Adequate resting analgesia</li> <li>• Decreased perioperative swelling and lack of incisional drainage</li> </ul> |

Continued



## Femoral Head and Neck Osteotomy—cont'd

| Phase                               | Expected Timeframe*                 | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Outcome Assessment Measures                                                                                                                                                                                                      | Criteria for Movement to Next Phase                                                                                                                                                                                               |
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| Early Weight-Bearing                | 72 hours to two weeks postoperative | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• PROM and flexion/extension hip stretches of operated limb (10-15 reps BID-TID). Bicycling and flexor reflex exercises</li> <li>• Slow, controlled walking on a land treadmill, 5-10 min including mild incline settings to encourage hip extension and target gluteal muscles</li> <li>• Balance exercises on a soft foam pad or bi-directional balance board for weight-bearing BID-TID</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy before activity (10-15 min BID-TID, NOT within 72 hours after surgery or if CS of acute inflammation are still present)</li> <li>• Therapeutic ultrasound SID</li> <li>• Massage</li> <li>• Therapeutic laser therapy PRN</li> <li>• Cryotherapy (15-20 min BID) following exercises</li> </ul> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• PROM and flexion/extension hip stretches of operated limb (10-15 reps BID-TID)</li> <li>• Slow, controlled leash walking, 5-20 min including mild inclines to encourage hip extension and target gluteal muscles</li> <li>• Balance exercises on a soft foam pad for weight-bearing BID-TID</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs—PRN</li> <li>• Opioids—PRN</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy (10-15 min BID-TID, prior to exercises, NOT within 72 hours after surgery or if CS of acute inflammation are still present)</li> <li>• Cryotherapy (15-20 min BID) following exercises</li> </ul> | <p>Goniometry—Hip ROM and other joints if applicable</p> <ul style="list-style-type: none"> <li>• Response to activity and subjective pain level</li> <li>• Lameness score at a stance and walk</li> <li>• Weight</li> </ul>     | <ul style="list-style-type: none"> <li>• Consistent partial weight-bearing on operated limb on all strides at a walk</li> <li>• Minimal pain with light activities</li> <li>• Incisional healing without complications</li> </ul> |
| Consistent Weight-Bearing at a Walk | Two to four weeks postoperative     | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• PROM and flexion/extension hip stretches of operated limb (10-15 reps SID-BID)</li> <li>• Controlled walking on a land treadmill, 10-15 min with increased incline angle and speed SID</li> <li>• Balance exercises on an inflatable disk or 360 degree wobble board for weight-bearing 5 min BID-TID</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• PROM and flexion/extension hip stretches of operated limb (10-15 reps BID-TID)</li> <li>• Leash walks 15-20 min including 5-10 min of inclines</li> <li>• Balance exercises on an inflatable disk for weight-bearing BID-TID</li> <li>• Sit-to-stand exercises 5-10 reps BID</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Goniometry—Hip ROM and other joints if applicable</p> <ul style="list-style-type: none"> <li>• Re-evaluate thigh muscle girth at 3-4 weeks postoperative</li> <li>• Response to activity and subjective pain level</li> </ul> | <ul style="list-style-type: none"> <li>• Consistent weight-bearing on operated limb on all strides at a walk, consistent partial weight-bearing at a trot</li> </ul>                                                              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                       |                                                                    |                                             |
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| <p>• Encourage increased weight-bearing on operated limb (examples: initiating dancing exercises as tolerated 5 min BID-TID, applying weight on operated limb at 3-5% body weight or syringe cap under contralateral foot)</p> <p>• Sit-to-stand exercises 5-10 reps BID</p> <p>• Aquatic therapy: UWTM walking 5-10 minutes once incision is sealed SID-BID</p> <p>• Swimming 3-5 days per week</p> <p>• Cavaletti Rails 5-10 reps BID</p> <p>Modalities</p> <p>• Heat therapy before activity</p> <p>• Therapeutic ultrasound PRN</p> <p>• Massage PRN</p> <p>• Therapeutic laser therapy PRN</p> <p>• Cryotherapy (15-20 min BID) following exercises</p>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                           | <p>• Light jogging 3-5 min per day</p> <p>• Stairs: 1 flight SID-BID</p> <p>Medications</p> <p>• NSAIDs—PRN</p> <p>Modalities</p> <p>• Heat therapy (10 min) before activity</p> <p>• Cryotherapy (15-20 min) following exercises</p> | <p>• Lameness score at a stance, walk and trot</p> <p>• Weight</p> | <p>• Minimal pain with light activities</p> |
| <p>Consistent Weight-Bearing at a Trot</p> <p>Five to eight weeks postoperative</p> <p>Therapeutic Exercises</p> <p>• PROM and flexion/extension hip stretches of operated limb PRN</p> <p>• Controlled walking or light jogging on a land treadmill, 15-20 min with increased incline angle and speed SID</p> <p>• Balance exercises on an inflatable disk or 360 degree wobble board for weight-bearing 10 min BID-TID</p> <p>• Sit-to-stand exercises 10-20 reps BID</p> <p>• Aquatic therapy: UWTM walking 15-30 minutes SID-BID</p> <p>• Swimming 2-5 days per week</p> <p>Modalities</p> <p>• Heat therapy PRN before activity</p> <p>• Cryotherapy PRN after exercises</p> | <p>Therapeutic Exercises</p> <p>• PROM and flexion/extension hip stretches of operated limb PRN</p> <p>• Leash walks 20-30 min including up to 10-15 min of inclines, may use weights on affected limb or pulled with a harness as tolerated/required</p> <p>• Incorporate challenging surfaces to walks, eg snow, sand, when possible</p> <p>• Controlled ball-playing with gradually increasing times and distances</p> <p>• Sit-to-stand exercises 10-20 reps BID</p> <p>• Light jogging 3-5 min per day</p> <p>• Stairs: 2-4 flights SID-BID</p> <p>Medications</p> <p>• NSAIDs—PRN</p> | <p>• Goniometry—Hip ROM and other joints if applicable</p> <p>• Re-evaluate thigh muscle girth at 7-8 weeks postoperative</p> <p>• Response to activity and subjective pain level</p> <p>• Lameness score at a stance, walk, and trot</p> <p>• Weight</p> | <p>• Consistent weight-bearing on operated limb at a trot</p> <p>• Minimal to no pain with moderate to extensive activities</p>                                                                                                       |                                                                    |                                             |

Continued

## Femoral Head and Neck Osteotomy—cont'd

| Phase                                                  | Expected<br>Timeframe*                    | Rehabilitation Clinic Program                                                                              | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Outcome<br>Assessment<br>Measures                                                                                                                                                                                                                                              | Criteria for<br>Movement to<br>Next Phase                                                                                                                                                              |
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| Trotting at Speed<br>with Minimal<br>to No<br>Lameness | Nine weeks<br>postoperative<br>and beyond | Aquatic therapy as desired; otherwise<br>exercises may be continued as part of<br>a Home Exercise Program. | Modalities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                        |
|                                                        |                                           |                                                                                                            | <ul style="list-style-type: none"> <li>Heat therapy PRN before activity</li> <li>Cryotherapy PRN after exercises</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                        |
|                                                        |                                           |                                                                                                            | Therapeutic Exercises <ul style="list-style-type: none"> <li>PROM for operated limb PRN</li> <li>Leash walks at times and distances tolerated, including fast walks up inclined surfaces</li> <li>Sit-to-stand exercises 20-30 reps as needed</li> <li>Jogging: working up from 10-15 minutes per day</li> <li>Stairs: walking and trotting, increasing number of flights as tolerated</li> <li>Swimming or walking in mid-to-upper-thigh-level water</li> <li>Ball-playing with gradually increasing times and distances, becoming more vigorous over time</li> </ul> | <ul style="list-style-type: none"> <li>Goniometry every 3-4 weeks if needed</li> <li>Re-evaluate thigh muscle girth every 3-4 weeks PRN</li> <li>Response to activity and subjective pain level</li> <li>Lameness score at a stance, walk, and trot</li> <li>Weight</li> </ul> | <ul style="list-style-type: none"> <li>Consistent weight-bearing on operated limb at a trot; permanent mild gait deficits may persist</li> <li>Minimal to no pain with extensive activities</li> </ul> |
|                                                        |                                           |                                                                                                            | Medications <ul style="list-style-type: none"> <li>NSAIDs—PRN</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                        |

\*This protocol is for use by licensed personnel who have been trained in canine rehabilitation. The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care. Patients receiving femoral head and neck surgery as a result of trauma, such as fracture of the acetabulum, generally require longer to recover than those receiving surgery as a result of severe hip dysplasia.

## Hip Luxation—Craniodorsal, Closed Reduction

| Phase                                                                                                       | Expected Timeframe*                   | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outcome Assessment Measures                                                                                                                                                                                                                                                                                                                  | Criteria for Movement to Next Phase                                                                                                                                                                                                           |
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| Immobilization Goals<br>Minimize post-procedural pain<br>Maintain coxofemoral joint in coaptation/reduction | Immediately to 2 weeks post-reduction | <p><i>The patient's affected pelvic limb should be placed in an Ehmer sling for 2 weeks after closed reduction, to maintain the limb abducted and internally rotated at the hip joint. This is necessary in hopes of maintaining hip reduction through early healing of joint capsule and periarticular soft tissues.</i></p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, controlled leash walks with Ehmer sling in place, 5 minutes only to go outside</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Cryotherapy at coxofemoral joint for 48-72 hours after closed reduction, 20 min TID</li> <li>• Heat therapy at coxofemoral joint (only after acute inflammation has subsided, at least 72 hours after reduction) 10 min BID-TID</li> <li>• Massage of lumbar spinal and accessible thigh muscles (while limb is immobilized)</li> <li>• TENS across coxofemoral joint 20 min BID-TID</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs/other analgesics</li> </ul> | <p><i>The patient's affected pelvic limb should be placed in an Ehmer sling for 2 weeks after closed reduction, to maintain the limb abducted and internally rotated at the hip joint. This is necessary in hopes of maintaining hip reduction through early healing of joint capsule and periarticular soft tissues.</i></p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, controlled leash walks with Ehmer sling in place, 5 minutes only to go outside</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Cryotherapy at coxofemoral joint for 48-72 hours after closed reduction, 20 min TID</li> <li>• Heat therapy at coxofemoral joint (only after acute inflammation has subsided, at least 72 hours after reduction) 10 min BID-TID</li> <li>• Massage of lumbar spinal and accessible thigh muscles (while limb is immobilized)</li> <li>• NSAIDs/other analgesics</li> </ul> | <ul style="list-style-type: none"> <li>• Morbidity associated with the sling apparatus</li> <li>• Range of motion of pelvic limb joints after immobilization period</li> <li>• Thigh circumference of affected limb after immobilization period</li> <li>• Degree of weight-bearing after immobilization period</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Well-maintained hip reduction with movement within the sagittal plane, indicating appropriate capsular/soft tissue healing under way</li> <li>• Adequate analgesia after closed reduction</li> </ul> |

Continued

## Hip Luxation—Craniodorsal, Closed Reduction\*\*—cont'd

| Phase                                                                                                                                                                                                         | Expected Timeframe*              | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Outcome Assessment Measures                                                                                                                                                                                                                                                                                                                          | Criteria for Movement to Next Phase                                                                                                                                                                                                                                                                                           |
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| Early Weight-Bearing Goals<br>Minimize pain with early activity<br>Improve weight-bearing of affected limb<br>Maintain hip in reduction with mild activity<br>Improve range of motion<br>Increase muscle mass | Two to four weeks post-reduction | <p>Therapeutic Exercises</p> <p><i>Care must be taken with all exercises to ensure external rotation and adduction of the hip do not occur at this stage.</i></p> <ul style="list-style-type: none"> <li>• Slow, controlled leash walks, 5 min TID-QID on flat surface or mild incline, support as needed</li> <li>• PROM and very gentle flexion/extension stretches of affected hip and other joints as needed 10-15 reps SID-BID. <i>It is critical to maintain movement in the sagittal plane</i></li> <li>• Aquatic Therapy: Underwater Treadmill (UWTM) walking 5 working up to 10 min SID-BID at a slow, steady pace with support as needed, beginning 4 weeks after reduction</li> <li>• Balance on a bi-directional rocker board with hindlimbs at a wide base (in slight abduction), 5-10 min SID-BID</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy at coxofemoral joint and thigh muscles 10 min BID-TID before exercise</li> <li>• Massage of lumbar spinal and thigh muscles</li> <li>• Therapeutic ultrasound (continuous) to coxofemoral joint SID</li> <li>• Therapeutic laser hip PRN</li> <li>• Cryotherapy at coxofemoral joint 20 min PRN after exercise</li> <li>• NMES affected thigh muscles 15 min EOD-SID PRN</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs/other analgesics PRN</li> </ul> | <p>Therapeutic Exercises</p> <p><i>Care must be taken with all exercises to ensure external rotation and adduction of the hip do not occur at this stage.</i></p> <ul style="list-style-type: none"> <li>• Slow, controlled leash walks, 5 min TID-QID on flat surface or mild incline, support as needed</li> <li>• PROM of affected hip and other joints as needed (10-15 reps SID-BID. <i>It is critical to maintain movement in the sagittal plane</i>)</li> <li>• Balance on a semi-firm surface such as a sofa cushion with hindlimbs at a wide base (in slight abduction), 5-10 min SID-BID</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy at coxofemoral joint and thigh muscles 10 min BID-TID before exercise</li> <li>• Massage—effleurage</li> <li>• Cryotherapy at coxofemoral joint 20 min PRN after exercise</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs/other analgesics PRN</li> </ul> | <ul style="list-style-type: none"> <li>• Goniometry—Hip ROM (flexion and extension only) and other joints if applicable</li> <li>• Response to light activity and subjective pain level</li> <li>• Thigh muscle circumference (evaluate at 2 and 4 weeks post reduction)</li> <li>• Lameness score at a stance and walk</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Consistent partial weight-bearing on affected limb on all strides at a walk</li> <li>• Minimal pain with light activities</li> <li>• Maintenance of hip reduction at current activity level</li> <li>• Hip PROM: &lt;60 degrees flexion and &gt;130-140 degrees extension</li> </ul> |

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| Consistent Weight-Bearing at a Walk     | Four to eight weeks post-reduction | Therapeutic Exercises<br><i>Care must be taken with all exercises to ensure external rotation and adduction of the hip do not occur at this stage.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Therapeutic Exercises<br><i>Care must be taken with all exercises to ensure external rotation and adduction of the hip do not occur at this stage.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Therapeutic Exercises<br><i>Care must be taken with all exercises to ensure external rotation and adduction of the hip do not occur at this stage.</i>                                                                         | Goniometry—<br>Hip ROM and other joints if applicable                                                                                                       | Consistent weight-bearing on affected limb on all strides at a walk, early weight bearing at a trot |
| Goals                                   |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                |                                                                                                                                                             |                                                                                                     |
| Minimize pain with increased activity   |                                    | <ul style="list-style-type: none"><li>Slow, controlled leash walks, 10-15 working up to 20 min TID-QID on flat surface or mild incline</li><li>PROM, bicycling, and flexion/extension stretches of affected hip and other joints as needed 10-15 reps SID-BID. <i>It is critical to maintain movement in the sagittal plane</i></li><li>Land Treadmill: 5 minutes SID walking on flat surface working up to slight inclines</li><li>Aquatic Therapy: Underwater Treadmill (UWTM) 10-20 min SID-BID at a slow, steady pace with support as needed</li><li>Balance on a 360-degree balance board with hindlimbs at a normal or slightly increased width (never adducted) 5-10 min SID-BID</li><li>Cavaletti Rails 5-10 laps BID</li><li>Sit-to-Stand 5-10 reps BID</li><li>Stair climbing 1-2 flights BID</li></ul> <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i> | <ul style="list-style-type: none"><li>Slow, controlled leash walks, 10-15 working up to 20 min TID-QID on flat surface or mild incline</li><li>PROM, bicycling and flexion/extension stretches of affected hip and other joints as needed 10-15 reps SID-BID. <i>It is critical to maintain movement in the sagittal plane</i></li><li>Balance on a moderately unstable surface with hindlimbs at a normal or slightly increased width (never adducted) 5-10 min SID-BID</li><li>Sit-to-stand 5-10 reps BID</li><li>Stair climbing 1-2 flights BID</li><li>Light jogging 1-2 min per day</li></ul> <i>Caution: Choose only 2-4 exercises per session. Doing all exercises may result in over-exertion and pain early in the program</i> | <ul style="list-style-type: none"><li>Thigh muscle circumference (re-evaluate every 2 weeks)</li><li>Response to activity and subjective pain level</li><li>Lameness score at a stance, walk and trot</li><li>Weight</li></ul> | <ul style="list-style-type: none"><li>Minimal pain with moderate activities</li><li>Hip PROM: 50-55 degrees flexion and 140-150 degrees extension</li></ul> |                                                                                                     |
| Improve weight-bearing of affected limb |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                |                                                                                                                                                             |                                                                                                     |
| Maintain hip in reduction               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                |                                                                                                                                                             |                                                                                                     |
| Improve range of motion                 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                |                                                                                                                                                             |                                                                                                     |
| Increase muscle mass                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                |                                                                                                                                                             |                                                                                                     |
|                                         |                                    | <p>Modalities</p> <ul style="list-style-type: none"><li>Heat therapy at coxofemoral joint and thigh muscles 10 min BID-TID before exercise</li><li>Therapeutic ultrasound (continuous) to coxofemoral joint PRN</li><li>Therapeutic laser hip PRN</li><li>Cryotherapy at coxofemoral joint 20 min PRN after exercise</li></ul> <p>Medications</p> <ul style="list-style-type: none"><li>NSAIDs/other analgesics PRN</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Modalities</p> <ul style="list-style-type: none"><li>Heat therapy at coxofemoral joint and thigh muscles 10 min BID-TID before exercise</li><li>Cryotherapy at coxofemoral joint 20 min PRN after exercise</li></ul> <p>Medications</p> <ul style="list-style-type: none"><li>NSAIDs/other analgesics PRN</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                |                                                                                                                                                             |                                                                                                     |

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## Hip Luxation—Craniodorsal, Closed Reduction\*\*—cont'd

| Phase                                                 | Expected Timeframe*                  | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outcome Assessment Measures                                                                                                                                                                                                                                                                               | Criteria for Movement to Next Phase                                                                                                                                                                            |
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| Consistent Weight-Bearing at a Trot                   | Eight to twelve weeks post-reduction | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, controlled leash walks, 10-15 working up to 20 min TID-QID on flat surface or mild incline</li> <li>• PROM, bicycling and flexion/extension stretches of affected hip and other joints as needed 10-15 reps SID-BID. <i>It is critical to maintain movement in the sagittal plane</i></li> <li>• Aquatic Therapy: Underwater Treadmill (UWTM) 20-30 minutes at variable speeds</li> <li>• Land Treadmill: 5-10 minutes SID walking with inclines, working up to use of resistance bands and weights on affected limb as tolerated. Incorporate 1-2 minutes jogging.</li> <li>• Balance on a 360-degree balance board 10 min BID</li> <li>• "Limbo" or tunnel crawl 5-10 reps SID-BID</li> <li>• Cavalletti Rails 10-20 laps BID</li> <li>• Weave poles 5-10 laps of 6-8 poles BID, gradually decreasing distance and increasing speed of navigation</li> <li>• Sit-to-stand 10-20 reps BID</li> <li>• Stair climbing 3-5 flights BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i></p> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• PROM and flexion/extension hip stretches of affected limb PRN</li> <li>• Leash walks 20-30 min including up to 10-15 min of mild-moderate inclines, may use weights on affected limb or pulled with a harness as tolerated/required</li> <li>• Incorporate challenging surfaces to walks, eg snow, sand, across a foam mattress, when possible</li> <li>• 3-5 min BID</li> <li>• Controlled ball-playing with gradually increasing times and distances</li> <li>• Sit-to-stand 10-20 reps BID</li> <li>• Light jogging 3-5 min per day</li> <li>• Stair climbing: 3-5 flights SID-BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Doing all exercises may result in over-exertion and pain early in the program</i></p> | <ul style="list-style-type: none"> <li>• Goniometry—Hip ROM and other joints if applicable</li> <li>• Thigh muscle circumference (re-evaluate every 2 weeks)</li> <li>• Response to activity and subjective pain level</li> <li>• Lameness score at a stance, walk, and trot</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Consistent weight-bearing on affected limb on all strides at a trot</li> <li>• Minimal pain with moderate to intense activities</li> <li>• Hip PROM normal</li> </ul> |
| Minimal to no pain with intense activity              |                                      | <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy PRN before activity</li> <li>• Cryotherapy PRN after exercises</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy PRN before activity</li> <li>• Cryotherapy PRN after exercises</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                |
| Normal weight-bearing of affected limb at all strides |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                |
| Normal hip range of motion                            |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                |
| Maintain hip in reduction through all movements       |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                |
| Normal muscle mass                                    |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                |

\*This protocol is for use by licensed personnel who have been trained in canine rehabilitation. The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

\*\*Closed reduction is not indicated in cases of hip dysplasia, femoral head fracture, or significant osteoarthritis of the hip. Chronic luxation warrants evaluation of the cartilage. If concurrent injuries preclude limb use, surgery may be required for earlier return of hip function.

| Osteoarthritis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                             |
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| Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outcome Assessment Measures                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Passive Range of Motion within comfortable range of affected joint(s), 15-20 repetitions, SID-BID</li> <li>• Balance/weight shifting on surface with stability appropriate to the patient's capability 5-10 min BID</li> <li>• Controlled stair climbing 1-3 flights per session BID-TID (to add to program only as appropriate)</li> <li>• Sit-to-stand exercises (if pelvic limb(s) affected) 10-15 reps BID</li> <li>• Cavaletti Rails 10-15 reps BID</li> <li>• Treadmill Walking or Jogging Downhill (for OA involving pelvic limbs) or Uphill (for OA involving thoracic limbs)—5-10% decline walking at a slow-to-medium walking speed, 5 min SID-TID</li> <li>• Aquatic Therapy: Underwater Treadmill Walking 5-10 min working up to 20-30 min SID-BID depending on patient tolerance and muscle strength/endurance</li> <li>• Swimming 5 working up to 20 minutes 2-4 times per week</li> </ul> <p><i>The mainstay of exercises for osteoarthritic patients is low impact, frequent activity of shorter duration; activity should not be of sufficient duration to exacerbate the patient's lameness. No more than 2-4 exercises should be performed per session in order to prevent pain due to excessive activity. At least two days of rest each week are recommended, in which exercises should consist only of PROM, stretching, and short leash walks</i></p> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Passive Range of Motion within comfortable range of affected joint(s), 15-20 repetitions, SID-BID</li> <li>• Balance/weight shifting on surface with stability appropriate to the patient's capability 5-10 min BID</li> <li>• Controlled leash walks with mild inclines 10-30 minutes as tolerated TID-QID</li> <li>• Controlled stair or ramp climbing 1-3 flights per session BID-TID (to add to program only as appropriate)</li> <li>• Walking on the side of inclines with affected side higher 5 min BID</li> <li>• Sit-to-stand exercises (if pelvic limb(s) affected) 10-15 reps BID</li> <li>• Trotting on lead (as tolerated, to add to program only if appropriate) 2-5 min SID</li> </ul> <p><i>The mainstay of exercises for osteoarthritic patients is more frequent activity of shorter duration; activity should not be of sufficient duration to exacerbate the patient's lameness. No more than 2-4 exercises should be performed per session in order to prevent pain due to excessive activity. At least two days of rest each week are recommended, in which exercises should consist only of PROM, stretching, and short leash walks</i></p> | <ul style="list-style-type: none"> <li>• Goniometric range of motion measurements for affected joint(s)</li> <li>• Bilateral limb circumference</li> <li>• Force plate evaluation</li> <li>• Stance analysis</li> <li>• Subjective gait analysis (using tools such as VAS)</li> <li>• Individualized functional measures set for each patient at the beginning of treatment, ie:</li> </ul> |

Continued

## Osteoarthritis—cont'd

| Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Outcome Assessment Measures                                                                                                                                     |
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| <p>Modalities</p> <ul style="list-style-type: none"> <li>• Massage: effleurage and massage of peri-articular muscle groups of affected limbs: 10 minutes at the start of each session</li> <li>• Transcutaneous electrical stimulation across affected joint(s) for pain relief (15-20 min SID)</li> <li>• NMES for muscle strengthening (15 min SID PRN)</li> <li>• Therapeutic Laser 10 J/cm<sup>2</sup> to affected joint(s)</li> <li>• Extracorporeal shock wave treatment</li> <li>• Cryotherapy (15-20 min BID-TID after exercise sessions) <i>Cryotherapy may also be utilized prior to exercise and stretching to improve pain-free range of motion</i></li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• Injectable polysulfated glycosaminoglycan agents</li> <li>• Oral chondroprotective agents</li> <li>• Omega-3 Fatty Acids (EPA/DHA)</li> <li>• NSAIDs PRN, ideally no more often than every other day long term</li> <li>• Opioids/Acetaminophen/NMDA Antagonists</li> <li>• Intra-articular corticosteroids; or hyaluronic acid, platelet rich plasma, and/or stem cell therapy</li> </ul> <p>Other</p> <ul style="list-style-type: none"> <li>• Weight management or weight loss program as appropriate with goal of BCS 4/9 to 5/9</li> </ul> | <p>Modalities</p> <ul style="list-style-type: none"> <li>• Massage: effleurage and massage of peri-articular muscle groups of affected limbs: 10 minutes before PROM and exercises</li> <li>• Cryotherapy (15-20 min BID-TID after exercise sessions) <i>Cryotherapy may also be used prior to exercise and stretching to improve pain-free range of motion</i></li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• Injectable polysulfated glycosaminoglycan agents</li> <li>• Oral chondroprotective agents</li> <li>• Omega-3 Fatty Acids (EPA/DHA)</li> <li>• NSAIDs PRN, ideally no more often than every other day long term</li> <li>• Opioids/Acetaminophen/NMDA Antagonists</li> </ul> <p>Other</p> <ul style="list-style-type: none"> <li>• Weight management or weight loss program as appropriate with goal of BCS 4/9 to 5/9</li> </ul> | <ul style="list-style-type: none"> <li>• Difficulty rising from a sit or lie down position</li> <li>• Difficulty navigating stairs</li> <li>• Weight</li> </ul> |

This protocol is for use only by licensed personnel who have been trained in canine rehabilitation.

The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

Osteoarthritis is a disease characterized by exacerbation and remission. The rehabilitation program should be appropriately adjusted to accommodate the varying phases/degrees of this condition.

## Patellar Luxation Repair

| Phase                                                                                                                                                                      | Expected Timeframe*                      | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outcome Assessment Measures                                                                                                                                                                                                                                                                            | Criteria for Movement to Next Phase                                                                                                                                                                                                                                                     |
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| <b>Step 1</b><br>Postoperative to Early Toe-Touching at a Walk<br><u>Goals:</u><br>Pain control<br>Swelling and edema control<br>Maintain/improve stifle postoperative ROM | Immediate to 48-72 hours postoperatively | Therapeutic Exercises <ul style="list-style-type: none"> <li>• Slow leash walking only to go outside, no more than five minutes at a time</li> <li>• Gentle PROM affected stifle within easy comfortable range, 8-10 repetitions BID-TID</li> </ul> <i>NOTE: It is critical that motion is limited to the sagittal plane.</i><br>Modalities <ul style="list-style-type: none"> <li>• Gentle massage (effleurage, thigh muscles)</li> <li>• Transcutaneous electrical stimulation for pain relief (15-20 min SID-TID)</li> <li>• Cryotherapy to affected stifle (15-20 min BID-TID)</li> <li>• Therapeutic Laser SID</li> </ul> Medications <ul style="list-style-type: none"> <li>• NSAIDs/other analgesics</li> <li>• Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists</li> </ul> | Inpatient status preferred during this phase<br>If home: <ul style="list-style-type: none"> <li>• Therapeutic Exercises</li> <li>• Slow leash walking only to go outside, no more than five minutes at a time</li> </ul> Modalities <ul style="list-style-type: none"> <li>• Gentle massage (effleurage)</li> <li>• Cryotherapy to affected stifle (15-20 min BID-TID)</li> </ul> Medications <ul style="list-style-type: none"> <li>• NSAIDs/other analgesics</li> <li>• Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists</li> </ul> | <ul style="list-style-type: none"> <li>• Postoperative bilateral stifle ROM (other joints if applicable)—goniometry</li> <li>• Postoperative bilateral thigh circumference</li> <li>• Response to activity and subjective pain level</li> <li>• Gait evaluation at a walk</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Consistent toe-touching on operated limb</li> <li>• Adequate resting analgesia.</li> <li>• Decreased perioperative swelling and no incisional drainage.</li> <li>• Comfortable ROM 70-90 degrees flexion, 120-135 degrees extension</li> </ul> |
| <b>Step 2</b><br>Early Weight-Bearing<br><u>Goals:</u><br>Continue pain control<br>Reduce swelling<br>Improve stifle ROM                                                   | 48-72 hours to two weeks postoperatively | Therapeutic Exercises <ul style="list-style-type: none"> <li>• Slow, steady leash walking 5 working up to 10 minutes at a time with sling support</li> <li>• Gentle PROM affected stifle within easy comfortable range, 10 repetitions BID-TID. Include hip/tarsal PROM of operated limb. <i>NOTE: It is critical that motion is limited to the sagittal plane.</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Therapeutic Exercises <ul style="list-style-type: none"> <li>• Slow, steady leash walking 5 working up to 10 minutes at a time with sling support</li> <li>• Balance/weight-shifting exercises on a semi-firm surface 5-10 min BID-TID</li> </ul>                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• Response to activity and subjective pain level</li> <li>• Gait evaluation at walk and trot</li> <li>• Goniometry—Stifle ROM and other joints if applicable</li> </ul>                                                                                         | <ul style="list-style-type: none"> <li>• Consistent partial weight-bearing with each stride on operated limb</li> <li>• Minimal pain with light activities</li> </ul>                                                                                                                   |

Continued

## Patellar Luxation Repair—cont'd

| Phase                                                                                                                                                         | Expected Timeframe*               | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Home Program                                                                                                                                                                                                                                                                                                                                                       | Outcome Assessment Measures                                                                                                                                                                              | Criteria for Movement to Next Phase                                                                                                                                                                                         |
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| Increase toe-touching weight-bearing<br>Prevent/minimize muscle atrophy                                                                                       |                                   | <ul style="list-style-type: none"> <li>Balance/weight-shifting exercises on a bi-directional board or soft foam pad 5-10 min BID-TID</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>Heat therapy prior to PROM exercises 10 min, only if acute inflammation is absent</li> <li>Gentle massage (effleurage, thigh muscles) prior to PROM</li> <li>Transcutaneous electrical stimulation across stifle for pain relief (15-20 min SID-TID)</li> <li>NMES 15 min for affected quadriceps and hamstring muscle group strengthening SID</li> <li>Cryotherapy to affected stifle (15-20 min PRN)</li> <li>Therapeutic Laser SID</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>NSAIDs/other analgesics</li> <li>Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists</li> </ul> | <p>Modalities</p> <ul style="list-style-type: none"> <li>Gentle massage (effleurage)</li> <li>Cryotherapy to affected stifle (15-20 min PRN)</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>NSAIDs/other analgesics</li> <li>Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists</li> </ul> | <ul style="list-style-type: none"> <li>Thigh circumference at 2 weeks postoperatively</li> <li>Weight</li> </ul>                                                                                         | <ul style="list-style-type: none"> <li>Incisional healing without complications</li> <li>ROM 60-80 degrees flexion, 130-150 degrees extension</li> </ul>                                                                    |
| <p><u>Step 3</u></p> <p>Consistent Weight-Bearing at a Walk</p> <p><u>Goals:</u></p> <p>Achieve normal stifle ROM</p> <p>Improve weight-bearing at a walk</p> | Two to five weeks postoperatively | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>Slow consistent leash walking—add 5 min every week if progressing well BID-QID</li> <li>PROM all joints of affected pelvic limb 10 repetitions BID-TID (PRN). <i>Maintain motion in the sagittal plane.</i></li> <li>Balance/weight-shifting exercises on a 360-degree balance board/unstable surface 5-10 min BID-TID</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Weight loss program can be initiated at this phase if warranted (minimizes loss of lean tissue mass).</p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>PROM all joints of affected pelvic limb 10 repetitions BID-TID (PRN). <i>Maintain motion in the sagittal plane.</i></li> </ul>                                                    | <ul style="list-style-type: none"> <li>Response to activity and subjective pain level</li> <li>Gait evaluation at walk and trot</li> <li>Goniometry—Stifle ROM and other joints if applicable</li> </ul> | <ul style="list-style-type: none"> <li>Consistent weight-bearing on operated limb at walk</li> <li>Early partial weight-bearing of operated limb at a trot</li> <li>No evidence of pain with moderate activities</li> </ul> |

Early weight-bearing at a trot  
Improve operated limb muscle mass/strength

- Sit-to-stand exercises 5-10 reps BID. *Use a sling or begin with hindquarters on an elevated surface if necessary. Begin at 4 weeks*
  - Low Cavaletti rails 5-10 reps BID, beginning at 4 weeks
  - Aquatic Therapy: Underwater Treadmill (UWTM) Walking 5-10 min SID-BID at a slow, steady pace
  - Land Treadmill (LTM): 5-10 min SID-BID at a slow, steady pace at a level plane.  
*Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program*
- Modalities
- Heat therapy prior to PROM exercises 10 min
  - Gentle massage (effleurage, thigh muscles) prior to PROM
  - Transcutaneous electrical stimulation across stifle for pain relief (15-20 min SID-TID PRN)
  - NMES 15 min for affected quadriceps and hamstring muscle group strengthening PRN
  - Cryotherapy to affected stifle (15-20 min after exercises PRN)
- Medications
- NSAIDs PRN
  - Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists
- Modalities
- Heat therapy prior to PROM exercises 10 min
  - Gentle massage (effleurage, thigh muscles) prior to PROM
  - Cryotherapy to affected stifle (15-20 min after exercises PRN)
- Medications
- NSAIDs—PRN
  - Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists
- Slow consistent leash walking—add 5 min every week if progressing well BID-QID
- 5 minutes of gentle uphill inclines *added to walks*
- Sit-to-stand exercises 5-10 reps BID. *Use a sling or begin with hindquarters on an elevated surface if necessary*
- Modalities
- Heat therapy prior to PROM exercises 10 min
  - Gentle massage (effleurage, thigh muscles) prior to PROM
  - Cryotherapy to affected stifle (15-20 min after exercises PRN)
- Medications
- NSAIDs—PRN
  - Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists
- Reassess thigh circumference at 3 and 5 weeks postoperatively
- Weight
- Stifle PROM 40-60 degrees flexion, >150 degrees extension

*Continued*



## Patellar Luxation Repair—cont'd

| Phase                                                       | Expected<br>Timeframe*                    | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Outcome<br>Assessment<br>Measures                                                                                                                                                                                                                                                                                   | Criteria for<br>Movement to<br>Next Phase                                                                                                                                                                                                             |
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| Step 4<br>Consistent Partial<br>Weight-Bearing<br>at a Trot | Five to eight<br>weeks<br>postoperatively | Therapeutic Exercises <ul style="list-style-type: none"> <li>PROM/stretching all joints of affected pelvic limb 10 repetitions BID-TID (PRN).</li> </ul> <i>Maintain motion in the sagittal plane.</i> <ul style="list-style-type: none"> <li>Balance/weight-shifting exercises on a 360-degree balance board/unstable surface 5-10 min BID-TID</li> <li>Sit-to-stand exercises 10-15 reps BID</li> <li>Cavaletti rails 10-15 reps BID</li> <li>Weave poles (decreasing distance and increasing speed adjusted to performance) 10-15 reps BID</li> <li>Aquatic Therapy: Underwater Treadmill (UWTM) walking 15-20 min SID-BID</li> <li>Swimming 5-10 min SID</li> <li>Land Treadmill (LTM): 5-10 min SID-BID at a mild incline. Resistance band use (<i>in sagittal plane ONLY</i>) and leg weights as applicable for muscle mass/strength improvement in the operated limb</li> </ul> | Continue weight loss program if needed<br>Therapeutic Exercises <ul style="list-style-type: none"> <li>PROM/stretching all joints of affected pelvic limb 10 repetitions PRN.</li> </ul> <i>Maintain motion in the sagittal plane.</i> <ul style="list-style-type: none"> <li>Controlled leash walking—add 5 min every week if progressing well BID-QID</li> <li>5 minutes of moderate uphill inclines on walks</li> <li>5 minutes circling, increasing tightness of circles every 4-5 sessions <i>added to walks</i></li> <li>2-3 minutes trotting added to walks</li> <li>“Dancing” exercises 5-10 reps of 5-10-second intervals BID-TID if acceptable to owner</li> <li>Walking up and down stairs, 1-2 flights BID-TID</li> <li>Sit-to-stand exercises 10-15 reps BID</li> </ul> | <ul style="list-style-type: none"> <li>Response to moderate/heavy activity and subjective pain level</li> <li>Goniometry—Stifle ROM</li> <li>and other joints if applicable</li> <li>Gait evaluation at walk and trot</li> <li>Reassess thigh circumference at 7-8 weeks postoperatively</li> <li>Weight</li> </ul> | <ul style="list-style-type: none"> <li>Minimal to no lameness at a walk.</li> <li>Consistent partial weight-bearing at a trot</li> <li>of normal speed</li> <li>No evidence of pain with activities</li> <li>Normal stifle range of motion</li> </ul> |

*Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program*

*Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program*

#### Modalities

- Heat therapy prior to exercises 10 min PRN
- NMES 15 min for affected quadriceps and hamstring muscle group strengthening PRN
- Cryotherapy to affected stifle (15-20 min after exercises PRN)

#### Medications

- NSAIDs PRN
- Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists

#### Modalities

- Heat therapy prior to exercises 10 min PRN
- Cryotherapy to affected stifle (15-20 min after exercises PRN)

#### Medications

- NSAIDs—PRN
- Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists

#### Step 5

Trotting at Speed with Minimal to No Lameness

#### Goals:

Continue improving speed at a lope and slow gallop

Continue improving muscle mass, strength, and power

Continue improving stamina and endurance

#### Eight weeks postoperatively and beyond

- #### Therapeutic Exercises
- Walking up and down stairs, 3-5 flights BID-TID building reps weekly as tolerated
  - Land treadmill 10-15 minutes with inclines, adding 2-3 minutes of jogging and increasing jogging time weekly as tolerated. Add leg weights and/or resistance bands as needed for muscle mass/strength development.
  - Sit-to-stand exercises 15-20 reps BID
  - Cavaletti rails 10-15 reps BID, increasing height and decreasing distance as appropriate

#### Therapeutic Exercises

- Leash walks up to 30 min BID-TID
- Trotting up and down inclines, 3-5 min added to walks
- Trotting in circles and figure 8s, 3-5 min BID
- Controlled ball play, 5-10 min SID-BID
- Walking up and down stairs, 3-5 flights BID-TID building reps weekly as tolerated
- “Dancing” exercises 5-10 reps of 15-20-second intervals BID-TID if acceptable to owner

Progressions may be continued as long as lameness or pain with activity is not observed.

- Response to increasing activity and subjective pain level
  - Gait evaluation at walk and trot
  - Evaluate thigh circumference every 2 weeks until symmetry achieved
  - Weight
- If this occurs, decrease activity by 50% for 1 week and increase again by 10-20% weekly.

*Continued*

## Patellar Luxation Repair—cont'd

| Phase                                    | Expected Timeframe* | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Outcome Assessment Measures | Criteria for Movement to Next Phase |
|------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------|
| Return to as normal activity as possible |                     | <ul style="list-style-type: none"> <li>Weave poles (decreasing distance and increasing speed adjusted to performance) 15-20 reps BID</li> <li>Aquatic therapy—UWTM for conditioning, consider progressively increasing speed and use of jets when available</li> <li>Swimming 10-15 minutes 3-4 times per week</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> <p>Modalities</p> <ul style="list-style-type: none"> <li>Heat therapy prior to exercises 10 min PRN</li> <li>Cryotherapy to affected stifle (15-20 min after exercises PRN)</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>NSAIDs PRN</li> <li>Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists</li> </ul> | <ul style="list-style-type: none"> <li>Sit-to-stand exercises 15-20 reps BID</li> </ul> <p><i>Caution: Choose only 2-3 exercises per session. Doing all exercises may result in lameness and pain early in the program</i></p> <p>Modalities</p> <ul style="list-style-type: none"> <li>Heat therapy prior to exercises 10 min PRN</li> <li>Cryotherapy to affected stifle (15-20 min after exercises PRN)</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>NSAIDs PRN</li> <li>Oral chondroprotective agents/injectable polysulfated glycosaminoglycans if patellofemoral OA exists</li> </ul> |                             |                                     |

This protocol is for use only by licensed personnel who have been trained in canine rehabilitation.

\*The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

## Thoracolumbar IVDE (postoperative)

| Phase                                           | Expected Timeframe*                          | Rehabilitation Clinic Program                                                                                                                                                         | Home Program                                                                                                                                                                          | Outcome Assessment Measures                                                                | Criteria for Movement to Next Phase                                                                 |
|-------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Rest                                            | Immediate to five-seven days postoperatively | Therapeutic Exercises                                                                                                                                                                 | Therapeutic Exercises                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Subjective pain level</li> </ul>                  | <ul style="list-style-type: none"> <li>• Postoperative comfort</li> </ul>                           |
| Goals:                                          |                                              | <ul style="list-style-type: none"> <li>• Assisted standing with abdominal support as needed on a stable, high-traction surface (<i>beginning 24 hours postoperatively</i>)</li> </ul> | <ul style="list-style-type: none"> <li>• Assisted standing with abdominal support as needed on a stable, high-traction surface (<i>beginning 24 hours postoperatively</i>)</li> </ul> | <ul style="list-style-type: none"> <li>• Hindlimb muscle tone</li> </ul>                   | <ul style="list-style-type: none"> <li>• Incisional integrity maintained</li> </ul>                 |
| Pain control                                    |                                              | <ul style="list-style-type: none"> <li>• PROM with bicycling movements all joints of pelvic limbs (10-15 reps BID-TID)</li> </ul>                                                     | <ul style="list-style-type: none"> <li>• PROM with bicycling movements all joints of pelvic limbs (10-15 reps BID-TID)</li> </ul>                                                     | <ul style="list-style-type: none"> <li>• Degree of support required in standing</li> </ul> | <ul style="list-style-type: none"> <li>• Standing in weight-bearing with minimal support</li> </ul> |
| Appreciable hindlimb tone                       |                                              | Modalities                                                                                                                                                                            | Modalities                                                                                                                                                                            |                                                                                            |                                                                                                     |
| Weight-bearing in standing with minimal support |                                              | <ul style="list-style-type: none"> <li>• TENS, premodulated or interferential electrical stimulation across incision site, 15-20 min SID</li> </ul>                                   | <ul style="list-style-type: none"> <li>• Cryotherapy at thoracolumbar incision site, 20 min BID-TID</li> </ul>                                                                        |                                                                                            |                                                                                                     |
| Minimize hindlimb muscle atrophy                |                                              | <ul style="list-style-type: none"> <li>• NMES to hindlimb musculature for contraction 15 min SID</li> </ul>                                                                           | <ul style="list-style-type: none"> <li>• Effleurage BID-TID</li> </ul>                                                                                                                |                                                                                            |                                                                                                     |
| Independent urination and defecation            |                                              | <ul style="list-style-type: none"> <li>• Cryotherapy at thoracolumbar incision site, 20 min BID-TID</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>• NSAIDs (if corticosteroids are not used)/opioids</li> </ul>                                                                                  |                                                                                            |                                                                                                     |
|                                                 |                                              | <ul style="list-style-type: none"> <li>• Gentle massage of hindlimb musculature +/- tapotement BID as tolerated</li> </ul>                                                            | <ul style="list-style-type: none"> <li>• NMDA receptor antagonists</li> </ul>                                                                                                         |                                                                                            |                                                                                                     |
|                                                 |                                              | <ul style="list-style-type: none"> <li>• Therapeutic laser therapy of epaxial musculature SID PRN</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>• Muscle relaxants BID-TID PRN</li> </ul>                                                                                                      |                                                                                            |                                                                                                     |
|                                                 |                                              | Other                                                                                                                                                                                 | Other                                                                                                                                                                                 |                                                                                            |                                                                                                     |
|                                                 |                                              | <ul style="list-style-type: none"> <li>• Bladder expression TID-QID PRN</li> </ul>                                                                                                    | <ul style="list-style-type: none"> <li>• Bladder expression TID-QID PRN</li> </ul>                                                                                                    |                                                                                            |                                                                                                     |
|                                                 |                                              | <ul style="list-style-type: none"> <li>• Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers</li> </ul>                                                     | <ul style="list-style-type: none"> <li>• Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers</li> </ul>                                                     |                                                                                            |                                                                                                     |

Continued

## Thoracolumbar IVDE (Postoperative)—cont'd

| Phase                                                                                                                                                                             | Expected Timeframe*              | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Home Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Outcome Assessment Measures                                                                                                                                                                                                                                    | Criteria for Movement to Next Phase                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Weight Support to Initial Motor Function                                                                                                                                          | One to two weeks postoperatively | Therapeutic Exercises <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting/rhythmic stabilization, gradually decreasing support (eg physioball beneath abdomen, standing in water)</li> <li>PROM with bicycling movements all joints of pelvic limbs (10-15 reps BID-TID)</li> <li>Aquatic therapy: Swimming 2-5 minutes working up to 10 minutes with supported standing during breaks</li> <li>Aquatic therapy: Underwater Treadmill (UWTM) walking, begin at 3-5 min up to 10-12 min. Ensure incision is sealed before beginning aquatic therapy</li> <li>Turning exercises during standing with positive reinforcement held at various locations (concentrate efforts on the weaker side)</li> </ul> Modalities <ul style="list-style-type: none"> <li>Gentle massage of hindlimb musculature BID-TID</li> <li>NMES to hindlimb musculature for contraction 15 min EOD-SID PRN</li> </ul> | Therapeutic Exercises <ul style="list-style-type: none"> <li>Assisted standing and weight-shifting/rhythmic stabilization, gradually decreasing support (eg physioball beneath abdomen)</li> <li>PROM with bicycling movements all joints of pelvic limbs (10-15 reps BID-TID)</li> <li>Slow leash walking with sling support 5 minutes TID</li> </ul> Modalities <ul style="list-style-type: none"> <li>Gentle massage of hindlimb musculature BID-TID</li> <li>Cryotherapy at thoracolumbar incision site, 20 min PRN after exercise</li> </ul> Medications <ul style="list-style-type: none"> <li>NSAIDs (if corticosteroids are not used)/opioids</li> <li>NMDA receptor antagonists</li> <li>Other <ul style="list-style-type: none"> <li>Bladder expression TID-QID PRN</li> <li>Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Response to activity and subjective pain level</li> <li>Length of time with unsupported standing</li> <li>Number of steps in which gait patterning is used</li> <li>Hindlimb thigh circumference/muscle mass</li> </ul> | <ul style="list-style-type: none"> <li>Pelvic limb voluntary motor activity</li> <li>Minimal pain with light activities</li> <li>Unsupported standing</li> </ul> |
| Goals:<br>Continued pain control<br>Full weight-bearing in standing<br>Minimized hindlimb muscle atrophy<br>Movement of all four limbs at a slow, supported walk<br>Self-rotation |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                |                                                                                                                                                                  |

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| Initial Motor Function to Good Motor Function with Proprioceptive Deficits<br><u>Goals:</u><br>Improve proprioception during walking activity<br>Maintain normal pelvic limb ROM<br>Increase pelvic limb muscle mass<br>Maintenance of balance during turns and on a moderately unstable surface | Two to four weeks postoperatively | <div><ul style="list-style-type: none"><li>• NMES to epaxial muscles for contraction 15 min EOD-SID PRN</li><li>• Cryotherapy at thoracolumbar incision site, 20 min PRN after exercise</li></ul><p>Other</p><ul style="list-style-type: none"><li>• Bladder expression TID-QID PRN</li><li>• Use of soft bedding with rotation every 4–6 hours to prevent decubital ulcers</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  | <div><ul style="list-style-type: none"><li>• Response to activity and subjective pain level</li><li>• Length of time with unsupported standing</li><li>• Length of time walking in aquatic therapy and on dry land</li><li>• Frequency of steps in which appropriate proprioception is used</li><li>• Height of Cavaletti rails navigated</li><li>• Pelvic limb thigh circumference/muscle mass</li></ul></div> |  |  |  |
|                                                                                                                                                                                                                                                                                                  |                                   | <div><p>Therapeutic Exercises</p><ul style="list-style-type: none"><li>• Assisted standing and weight-shifting on a moderately unstable surface</li><li>• PROM with bicycling movements all joints of pelvic limbs (10-15 reps PRN based on ROM)</li><li>• Slow leash walking with sling available 10 minutes BID-TID</li><li>• Turning exercises during standing with positive reinforcement held at various locations (concentrate efforts on the weaker side)</li><li>• Turning exercises during walking: weave poles gradually decreasing in distance 5-10 reps SID-BID</li></ul><p>Modalities</p><ul style="list-style-type: none"><li>• Cryotherapy at thoracolumbar incision site, 20 min PRN after exercise</li></ul><p>Medications</p><ul style="list-style-type: none"><li>• NSAIDs PRN</li><li>• NMDA receptor antagonists</li></ul></div> |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |

Continued



## Thoracolumbar IVDE (postoperative)—cont'd

| Phase                                                                         | Expected Timeframe*         | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Home Program                                                                                                                                                                                                                                                                                                                                                                                               | Outcome Assessment Measures                                                                                                                                                                                                                      | Criteria for Movement to Next Phase                                                                                                                                                                     |
|-------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                               |                             | <ul style="list-style-type: none"> <li>• Low Cavaletti rails 5-10 reps SID-BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> <p>Modalities</p> <ul style="list-style-type: none"> <li>• NMES to hindlimb musculature for contraction 15 min EOD-SID PRN</li> <li>• NMES to epaxial muscles for contraction 15 min EOD-SID PRN</li> <li>• Cryotherapy at thoracolumbar incision site, 20 min PRN after exercise</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                         |
| Good motor function with mild proprioceptive deficits to near normal gait     | Four to six postoperatively | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Assisted standing and weight-shifting on a 360-degree balance board</li> <li>• Aquatic therapy: Swimming 20-30 minutes with supported standing during breaks</li> <li>• Aquatic therapy: Underwater Treadmill (UWTM) walking, 15-20 minutes</li> </ul>                                                                                                                                                                                                                                                                       | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Assisted standing and weight-shifting on an unstable surface</li> <li>• Turning exercises during walking: weave poles gradually decreasing in distance 5-10 reps SID-BID</li> <li>• Stair Climbing 1 working up to 5 flights SID-BID</li> <li>• Sit-to-stand exercises, rising from an elevated surface 5-10 reps SID-BID</li> </ul> | <ul style="list-style-type: none"> <li>• Response to activity and subjective pain level</li> <li>• Length of time with unsupported standing</li> <li>• Walking endurance</li> <li>• Tightness of turns in which balance is maintained</li> </ul> | <ul style="list-style-type: none"> <li>• Consistent pelvic limb proprioception at a walk</li> <li>• Minimal to no pain with extensive activities</li> <li>• Normal gait patterning at a trot</li> </ul> |
| Goals:<br>Consistent proprioception with moderate- to high-intensity activity |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                         |

|                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Pelvic limb muscle mass approaching normal</p> <p>Maintain balance during turns and negotiating obstacles while walking</p> | <ul style="list-style-type: none"> <li>• Gait patterning on a ground treadmill, 5-10 minutes with resistance band use PRN</li> <li>• Turning exercises during walking: weave poles gradually decreasing in distance 5-10 reps SID-BID</li> <li>• Cavaletti rails 5-10 reps SID-BID</li> <li>• Stair Climbing 1 working up to 5 flights SID-BID</li> <li>• Sit-to-stand exercises, rising from an elevated surface 5-10 reps SID-BID</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Cryotherapy at thoracolumbar incision site, 20 min PRN after exercises</li> </ul> | <ul style="list-style-type: none"> <li>• Leash walking 10-15 min BID-TID with 5 minutes inclines/declines</li> <li>Modalities</li> <li>• Cryotherapy at thoracolumbar incision site, 20 min PRN after exercise</li> <li>Medications</li> <li>• NSAIDs PRN</li> <li>• NMDA receptor antagonists</li> </ul>                        | <ul style="list-style-type: none"> <li>• Frequency of steps in which appropriate proprioception is used</li> <li>• Height of Cavaletti rails navigated</li> <li>• Pelvic limb thigh circumference/muscle mass</li> </ul>                                                                                                                   |
| <p>Near Normal Gait to Normal Gait</p> <p>Goals:<br/>Elimination of proprioceptive deficits during high-intensity activity</p> | <p>Six or more weeks postoperatively</p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Assisted standing and weight-shifting on a 360-degree balance board</li> <li>• Aquatic therapy: Swimming 20-30 minutes</li> <li>• Aquatic therapy: Underwater Treadmill (UWTM) walking, 20-30 minutes at variable speeds</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Assisted standing and weight-shifting on an unstable surface</li> <li>• Turning exercises with weave poles 10-20 reps SID-BID at a trot</li> <li>• Stair Climbing 5-10 repetitions SID-BID</li> <li>• Sit-to-stand exercises 10-20 reps SID-BID</li> </ul> | <ul style="list-style-type: none"> <li>• Response to activity and subjective pain level</li> <li>• Walking and trotting endurance</li> <li>• Tightness of turns in which balance is maintained</li> <li>• Minimal to no pain with extensive activities</li> <li>• Normal gait and proprioception at a trot and at faster speeds</li> </ul> |

Continued

## Thoracolumbar IVDE (postoperative)—cont'd

| Phase                                                                                                    | Expected Timeframe* | Rehabilitation Clinic Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Home Program                                                                                                                                                                                                            | Outcome Assessment Measures                                                                                                                                 | Criteria for Movement to Next Phase |
|----------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Normal pelvic limb muscle mass<br>Maintain balance during turns and negotiating obstacles while trotting |                     | <ul style="list-style-type: none"> <li>• Gait patterning on a ground treadmill, 5-10 minutes at a trot</li> <li>• Turning exercises with weave poles 10-20 reps</li> <li>• SID-BID at a trot</li> <li>• Cavaletti rails 10-20 reps</li> <li>• SID-BID at a trot</li> <li>• Stair Climbing 5-10 repetitions</li> <li>• Sit-to-stand exercises 10-20 reps</li> <li>• SID-BID</li> <li>• Controlled ball play</li> </ul> <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> | <ul style="list-style-type: none"> <li>• Controlled ball play</li> <li>• Leash walking 20-30 minutes</li> <li>• BID with 10 minutes variable inclines and 5-10 minutes working up to longer periods trotting</li> </ul> | <ul style="list-style-type: none"> <li>• Height and distance of Cavaletti rails navigated</li> <li>• Pelvic limb thigh circumference/muscle mass</li> </ul> |                                     |

This protocol is for use by licensed personnel who have been trained in canine rehabilitation.

\*The therapist must keep in mind that these timelines are intended as guidelines. Rate of recovery will vary widely among patients for reasons that include status of nociception and function prior to and immediately after surgery.

## Total Hip Replacement

| Phase                                                                                                                                               | Expected<br>Timeframe*               | Rehabilitation Clinic Program**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Home Program**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Outcome<br>Assessment<br>Measures                                                                                                                                                                                                                                                  | Criteria for<br>Movement to<br>Next Phase                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-Weight-Bearing to Toe-Touching<br><u>Goals:</u><br>Pain control<br>Swelling and edema control<br>Early limb use<br>Slow/minimize muscle atrophy | Immediate to 2 weeks postoperatively | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, gentle PROM for operated limb (8-10 reps flexion and extension BID). <i>Sagittal plane only—abduction and adduction of the hip must be STRICTLY AVOIDED</i></li> <li>• Slow leash walking with firm sling support, only to go outside (up to 5 minutes, TID-QID)</li> <li>• Balance exercises on a soft foam pad or bi-directional balance board for early toe-touching weight-bearing (begin at early toe-touching 3-5 days after surgery, use assisted sling support)</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Gentle massage around the operated hip, thigh, and lumbosacral regions, avoiding the incision</li> <li>• Transcutaneous electrical stimulation to hip for pain relief (15-20 min SID-BID)</li> <li>• Cryotherapy to hip (15-20 min TID for 48-72 hours then PRN after light activity)—first session immediately postoperatively</li> <li>• Therapeutic laser therapy SID</li> <li>• NMES—quadriceps and hamstrings muscle groups (15 min SID), begin 5-7 days postoperatively after acute inflammation has been addressed</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs</li> <li>• Opioids</li> <li>• NMDA Antagonists</li> </ul> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Slow, gentle PROM for stifle and tarsus of the operated limb (8-10 reps flexion and extension TID-QID)</li> <li>• Slow leash walking with firm sling support, only to go outside (up to 5 minutes, TID-QID)</li> <li>• Weight-shifting exercises on a stable surface for early toe-touching weight-bearing (begin at early toe-touching 3-5 days after surgery, use assisted sling support)</li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Effleurage, avoiding incision, BID-TID</li> <li>• Cryotherapy to hip (15-20 min TID for 48-72 hours then PRN after light activity)</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs</li> <li>• Opioids</li> <li>• NMDA Antagonists</li> </ul> | <ul style="list-style-type: none"> <li>• Postoperative hip sagittal plane PROM and other joints as applicable via goniometry</li> <li>• Postoperative bilateral thigh circumference</li> <li>• Response to minimal activity and subjective pain level</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Early toe-touching</li> <li>• Adequate resting analgesia</li> <li>• Decreased perioperative swelling and no incisional drainage.</li> </ul> |

## Total Hip Replacement—cont'd

| Phase                                                                                                                                                                                                | Expected Timeframe*        | Rehabilitation Clinic Program**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Home Program**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outcome Assessment Measures                                                                                                                                                                                                                                               | Criteria for Movement to Next Phase                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Toe-Touching to Consistent Partial Weight-Bearing at a Walk<br><br>Goals:<br>Continued pain control<br>Prevention of further muscle atrophy, early strengthening<br>Improved limb function at a walk | 2-12 weeks postoperatively | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Stretching and PROM of the operated limb (8-10 reps flexion and extension BID before activities). <i>Sagittal plane only—abduction and adduction of the hip must be STRICTLY AVOIDED</i></li> <li>• Slow leash walking with firm sling support, 3-5 min working up to 10 min steady walking TID</li> <li>• Balance and weight-shifting exercises on a soft foam pad or bi-directional balance board (use assisted sling support)</li> <li>• Aquatic therapy: Underwater Treadmill (UWTM) walking: begin at 3-5 min working up to 10-15 min SID-BID. <i>Begin at 3-4 weeks postoperatively</i></li> <li>• Low Cavaletti Rails: 8-10 reps BID. <i>Begin at 4-6 weeks postoperatively, working up to higher rails by 8-12 weeks postoperatively</i></li> <li>• Sit-to-Stand with hindquarters on an elevated surface, 5-10 reps BID-TID. <i>Begin at 8-12 weeks postoperatively</i><br/><i>Caution: Choose only 1-2 exercises per session. Doing all exercises may result in lameness and pain early in the program</i></li> </ul> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Stretching and PROM with bicycling of the operated limb (8-10 reps flexion and extension BID-TID before activities)—<i>abduction and adduction of the hip must be STRICTLY AVOIDED</i></li> <li>• Slow leash walking with firm sling support, 3-5 min working up to 10 min steady walking TID</li> <li>• Balance and weight-shifting exercises on a semi-firm stable surface (use assisted sling support)</li> <li>• Flexor reflex exercises of operated hindlimb</li> <li>• Sit-to-Stand with hindquarters on an elevated surface, 5-10 reps BID-TID. <i>Begin at 8-12 weeks postoperatively</i><br/><i>Caution: Choose only 1-2 exercises per session. Doing all exercises may result in lameness and pain early in the program</i></li> </ul> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Cryotherapy to hip (15-20 min PRN after light activity)</li> </ul> | <ul style="list-style-type: none"> <li>• Goniometry—Hip ROM and other joints if applicable</li> <li>• Response to activity and subjective pain level</li> <li>• Bilateral thigh circumference</li> <li>• Lameness score at a stance and walk</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Consistent partial weight-bearing on operated limb on all strides at a walk</li> <li>• Minimal pain with light activities</li> <li>• Incisional healing without complications</li> </ul> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Modalities</p> <ul style="list-style-type: none"> <li>• Transcutaneous electrical stimulation to hip for pain relief (15-20 min SID-BID PRN)</li> <li>• Cryotherapy to hip (15-20 min PRN after light activity)</li> <li>• NMES—quadriceps and hamstrings muscle groups (15 min EOD-SID)</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs SID-PRN</li> <li>• Chondroprotective agents as needed for other joints</li> </ul> | <p>Consistent Weight-Bearing at a Walk to Early Trotting</p> <p>Goals:</p> <p>Muscle strengthening</p> <p>Normal limb function at a walk, improved function at faster gaits</p> <p>Near-normal hip PROM in flexion/extension</p> <p>12-16 weeks postoperatively</p> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Stretching of the operated limb (8-10 reps BID-TID before activities)</li> <li>• 3-5 min leash walk with 30 sec-1 min trotting as a warm-up</li> <li>• Land treadmill walking 5-10 min SID-BID on a flat surface, with resistance band applied in the sagittal plane to strengthen operated limb</li> <li>• Balance and weight-shifting exercises on a 360-degree balance board, 10 min SID-BID (have sling support available if needed)</li> <li>• Aquatic therapy: Underwater Treadmill (UWTM) walking: 15-20 min SID-BID</li> <li>• Swimming: 5-10 min working up to 20 min SID with breaks</li> <li>• Cavaletti Rails: 10-15 reps BID</li> <li>• Limbo crawling: 8-10 reps BID</li> <li>• Sit-to-Stand 10-15 reps BID-TID</li> <li>• Controlled stair walking on leash: 1-2 flights SID-BID</li> </ul> <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Stretching of the operated limb (8-10 reps BID-TID before activities).</li> <li>• Slow leash walking 15-20 min BID.</li> <li>• Incorporate 5 min light inclines and 3-5 min slow trotting on a flat surface</li> <li>• Balance and weight-shifting exercises on a semi-unstable surface (have sling support available if needed)</li> <li>• Sit-to-Stand 10-15 reps BID-TID</li> <li>• Controlled stair walking on leash: 1-2 flights SID-BID</li> </ul> <p>Caution: Choose only 2-3 exercises per session. Doing all exercises may result in lameness and pain early in the program</p> <p>Goniometry—</p> <ul style="list-style-type: none"> <li>• Hip ROM and other joints if applicable</li> <li>• Bilateral thigh muscle girth</li> <li>• Response to activity and subjective pain level</li> <li>• Lameness score at a stance, walk, and trot</li> <li>• Weight</li> </ul> <p>Consistent weight-bearing on operated limb on all strides at a walk, consistent partial weight bearing at a trot</p> <ul style="list-style-type: none"> <li>• Minimal pain with moderate activities</li> </ul> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Continued



## Total Hip Replacement—cont'd

| Phase                                                                                                                                              | Expected Timeframe*       | Rehabilitation Clinic Program**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Home Program**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Outcome Assessment Measures                                                                                                                                                                                                                                                                     | Criteria for Movement to Next Phase                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Early to Consistent Weight-Bearing at a Trot<br>Goals:<br>Hindlimb muscular symmetry<br>Normal limb function at a walk and trot<br>Normal hip PROM | >16 weeks postoperatively | <p><i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i></p> <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy to hip (10 min PRN with stretching before exercises)</li> <li>• Cryotherapy to hip (15-20 min PRN after exercises)</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs PRN</li> <li>• Chondroprotective agents as needed for other joints</li> </ul>                                                                                                                                                                        | <p>Modalities</p> <ul style="list-style-type: none"> <li>• Heat therapy to hip (10 min PRN with stretching before exercises)</li> <li>• Cryotherapy to hip (15-20 min PRN after exercises)</li> </ul> <p>Medications</p> <ul style="list-style-type: none"> <li>• NSAIDs PRN</li> <li>• Chondroprotective agents as needed for other joints</li> </ul>                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Goniometry—Hip ROM and other joints if applicable</li> <li>• Bilateral thigh muscle girth</li> <li>• Response to moderate-intense activity and subjective pain level</li> <li>• Lameness score at a stance, walk and trot</li> <li>• Weight</li> </ul> | <ul style="list-style-type: none"> <li>• Consistent weight-bearing on operated limb at all gaits</li> <li>• Minimal to no pain with moderate to extensive activities</li> <li>• Hindlimb muscle subjective symmetry</li> </ul> |
|                                                                                                                                                    |                           | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Stretching of the operated limb (8-10 reps PRN before activities)</li> <li>• 5 min leash walk with 1-2 min trotting as a warm-up</li> <li>• Land treadmill walking 10-15 min SID-BID on a flat surface, with resistance band applied in the sagittal plane and leg weights as needed for atrophy on operated limb</li> <li>• Balance and weight-shifting exercises on a 360-degree balance board, 10 min SID-BID</li> <li>• Aquatic therapy: Underwater Treadmill (UWTM) walking: 15-20 min SID-BID, variable speeds</li> <li>• Swimming: 20-30 min SID with breaks</li> <li>• Cavaletti Rails: 10-15 reps BID</li> <li>• Limbo crawling: 10-15 reps BID</li> </ul> | <p>Therapeutic Exercises</p> <ul style="list-style-type: none"> <li>• Stretching of the operated limb (8-10 reps PRN before activities)</li> <li>• Slow leash walking 15-20 min BID.</li> <li>• Incorporate 5-min moderate inclines and gradually increase slow trotting on a flat surface</li> <li>• Balance and weight-shifting exercises on a semi-unstable surface (leg weights as needed for atrophy on operated limb)</li> <li>• Sit-to-Stand 15-20 reps BID-TID</li> <li>• Controlled stair walking on leash: 3-5 flights SID-BID</li> </ul> |                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                |

- Sit-to-Stand 15-20 reps BID-TID
- Controlled stair walking on leash: 3-5 flights SID-BID  
*Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program.*  
*Activity should continue to be gradually increased based on each patient's individual progress. Jumping and sharp turns should not be allowed until return to full function, strength, and endurance are observed.*  
**Modalities**
  - Heat therapy to hip (10 min PRN with stretching before exercises)
  - Cryotherapy to hip (15-20 min PRN after exercises)**Medications**
  - NSAIDs PRN
- Controlled ball play, 5-10 min SID-BID  
*Caution: Choose only 2-3 exercises per session.*  
*Doing all exercises may result in lameness and pain early in the program.*  
*Activity should continue to be gradually increased based on each patient's individual progress. Jumping and sharp turns should not be allowed until return to full function, strength, and endurance are observed.*  
**Modalities**
  - Heat therapy to hip (10 min PRN with stretching before exercises)
  - Cryotherapy to hip (15-20 min PRN after exercises)**Medications**
  - NSAIDs PRN

\*This protocol is for use by licensed personnel who have been trained in canine rehabilitation. The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

\*\*Strict exercise restriction, with the exception of controlled rehabilitation exercises, should be maintained for 8-12 weeks following this procedure.



# 2

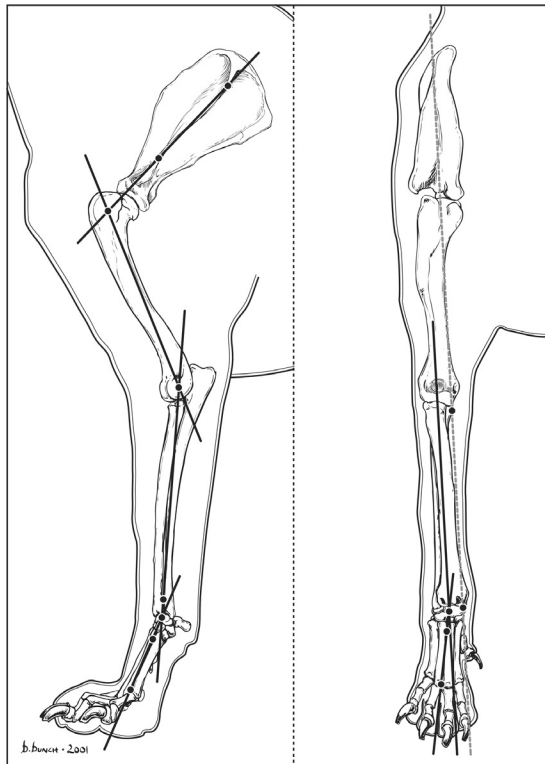
## APPENDIX

# Joint Motions and Ranges

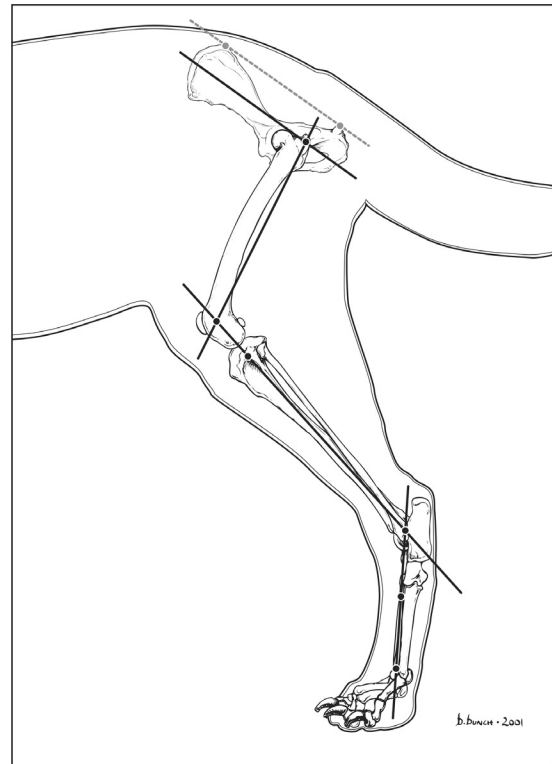
| Joints                               | Joint Motions                   | Normal Range of Motion for Dogs (degrees) |
|--------------------------------------|---------------------------------|-------------------------------------------|
| Forelimb Joints                      |                                 |                                           |
| Shoulder (including scapular motion) | Flexion                         | 30-60                                     |
|                                      | Extension                       | 160-170                                   |
|                                      | Abduction                       | 40-50                                     |
|                                      | Adduction                       | 40-50                                     |
|                                      | Internal rotation               | 40-50                                     |
|                                      | External rotation               | 40-50                                     |
| Elbow                                | Flexion                         | 20-40                                     |
|                                      | Extension                       | 160-170                                   |
|                                      | Hyperextension or overextension |                                           |
| Radioulnar                           | Pronation                       | 40-50                                     |
|                                      | Supination                      | 80-90                                     |
| Carpus (Wrist)                       | Flexion                         | 20-35                                     |
|                                      | Hyperextension                  | 190-200                                   |
|                                      | Radial or medial deviation      | 5-15                                      |
|                                      | Ulnar or lateral deviation      | 10-20                                     |
| Hip                                  | Flexion                         | 55                                        |
|                                      | Extension                       | 160-165                                   |
|                                      | Abduction with a flexed hip     | 120 (Stifle at 90)                        |
|                                      | Adduction with a flexed hip     | 65 (Stifle at 90)                         |
|                                      | Abduction with an extended hip  | 85                                        |
|                                      | Adduction with an extended hip  | 63                                        |
|                                      | Internal rotation               | 55                                        |
|                                      | External rotation               | 50                                        |
| Stifle (knee)                        | Flexion                         | 45                                        |
|                                      | Extension                       | 160-170                                   |
| Talocrural, tarsocrural, ankle       | Flexion                         | 40                                        |
|                                      | Extension                       | 170                                       |

NOTE: Presently, the tendency is to call the straight joint position 180° rather than 0° being established. Range of motion is expressed in degrees unless noted otherwise.

## Goniometry of the Forelimb and Hindlimb

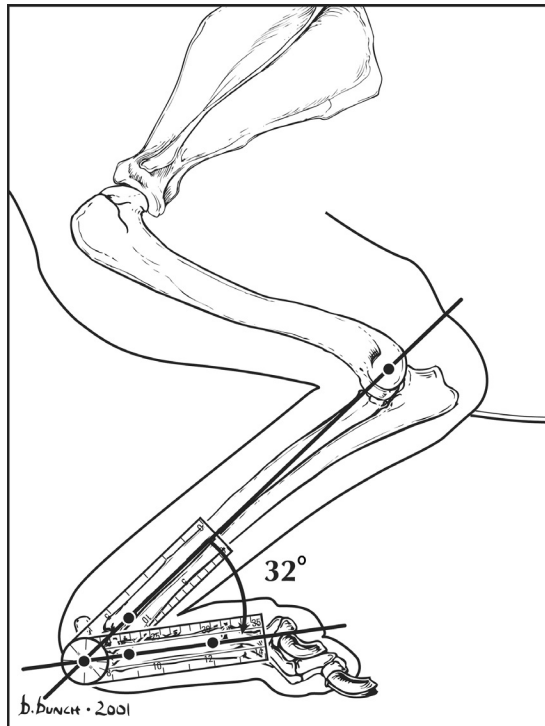


**Figure 1** Goniometry of the forelimb. Carpal flexion and extension are measured as the angles formed by the long axis of metacarpal bones III and IV and the line joining the cranial to caudal midpoint of the antebrachium at the level of the ulnar styloid process and the lateral humeral epicondyle. Elbow flexion and extension are measured as the angles formed by the line joining the cranial to caudal midpoint of the antebrachium at the level of the ulnar styloid process and the lateral humeral epicondyle and a line joining the lateral epicondyle to the point of insertion of the infraspinatus muscle on the greater tubercle of the humerus. Shoulder flexion and extension are measured as the angles formed by the line joining the lateral humeral epicondyle and the point of insertion of infraspinatus muscle and the spine of the scapula. Carpal varus and valgus are measured as the angles formed by the long axis of metacarpals III and IV and the long axis of the medial border of the radius.

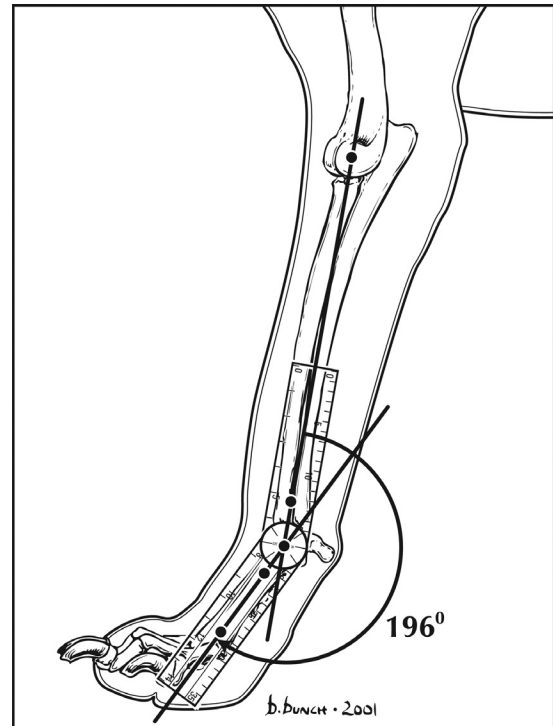


**Figure 2** Goniometry of the hindlimb. Tarsal flexion and extension are measured as the angles formed by the long axis of metatarsal bones III and IV and the long axis of the tibial shaft. Flexion and extension of the stifle joint are measured as the angles formed by the long axis of the tibial shaft and the line joining the lateral femoral epicondyle and greater trochanter. Hip joint flexion and extension are measured as the angles formed by the line joining the lateral femoral epicondyle of the femur and greater trochanter and a line joining the tuber sacrale and ischiadicum.

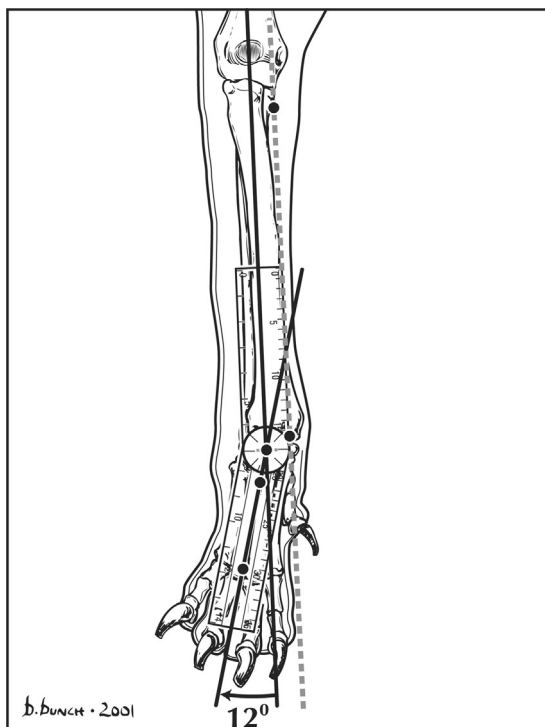
The figures and Table 1 used by permission from Jaegger G, Marcellin-Little DJ, Levine D: Reliability of goniometry in Labrador retrievers, *Am J Vet Res* 63: 979-986, 2002



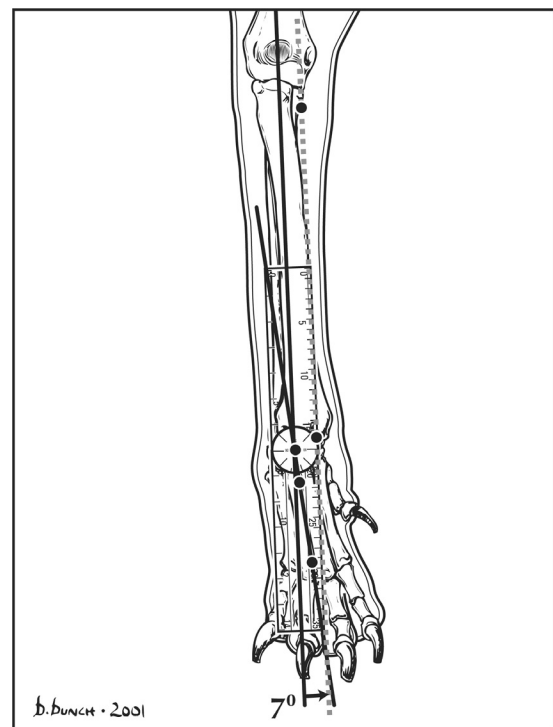
**Figure 3** Carpal flexion.



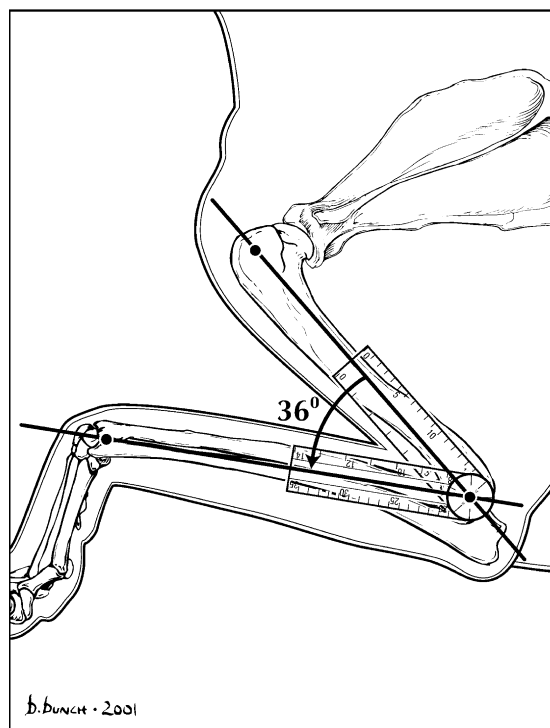
**Figure 4** Carpal extension.



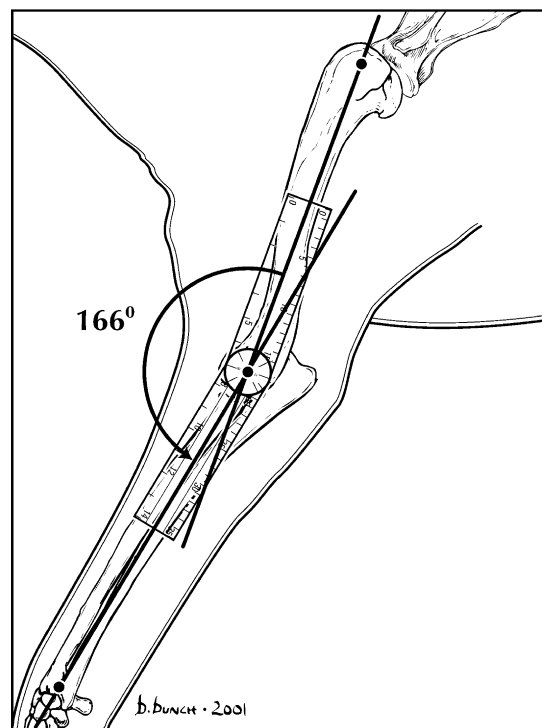
**Figure 5** Carpal valgus.



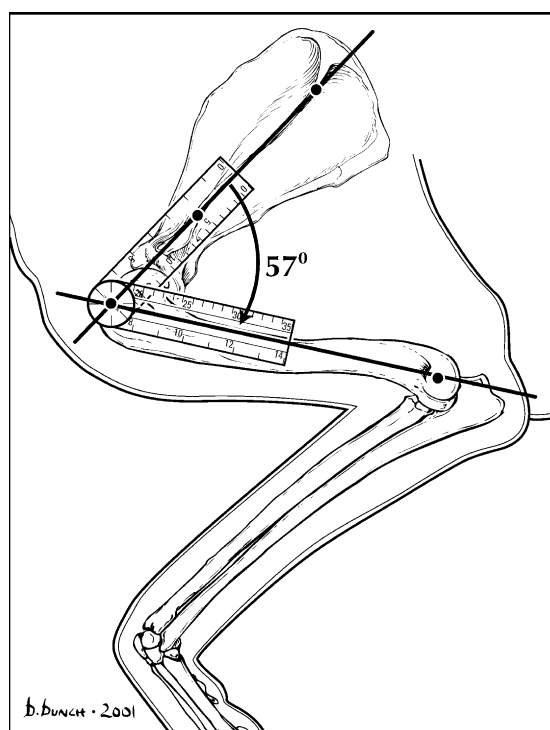
**Figure 6** Carpal varus.



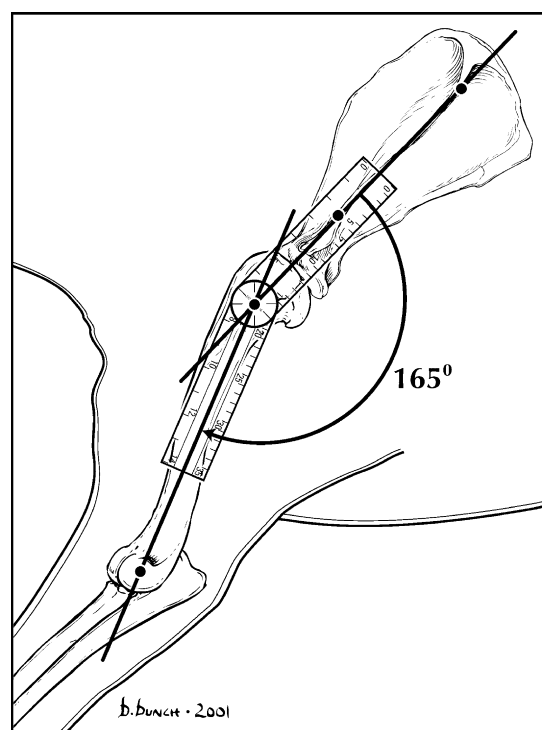
**Figure 7** Elbow flexion.



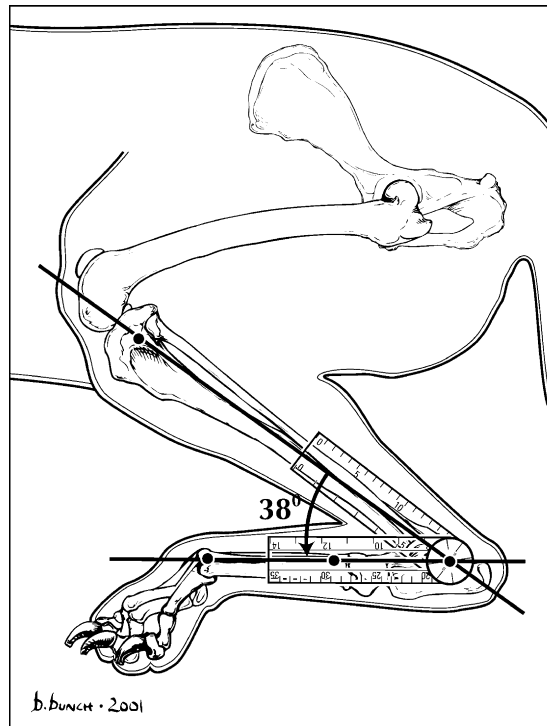
**Figure 8** Elbow extension.



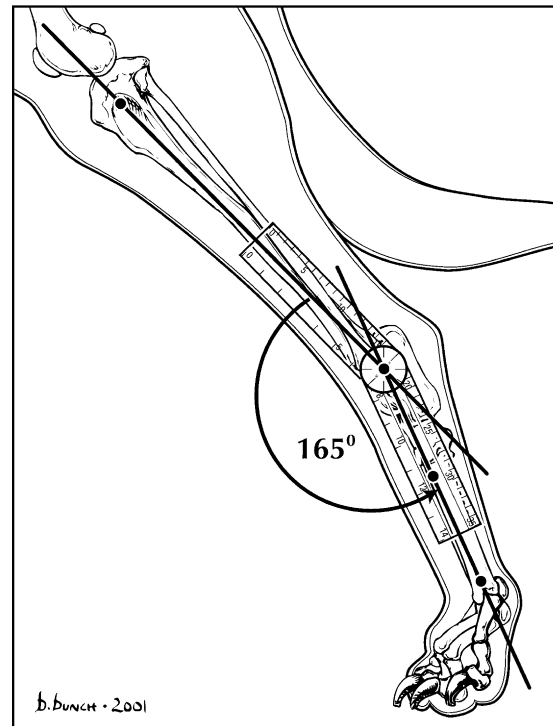
**Figure 9** Shoulder flexion.



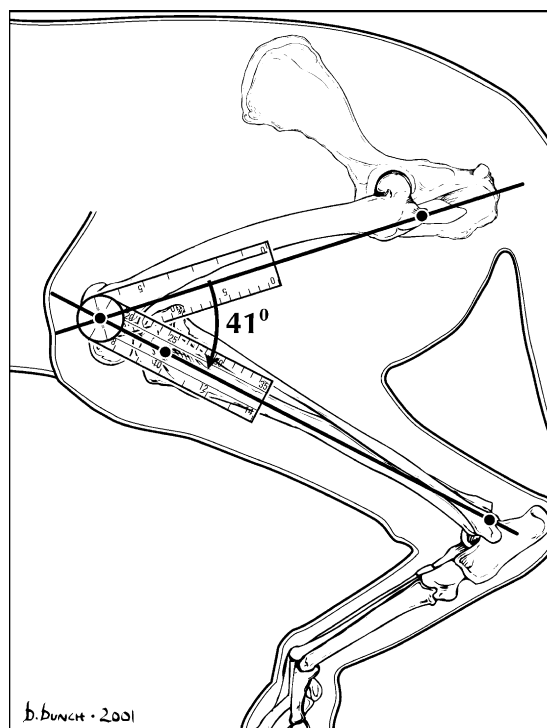
**Figure 10** Shoulder extension.



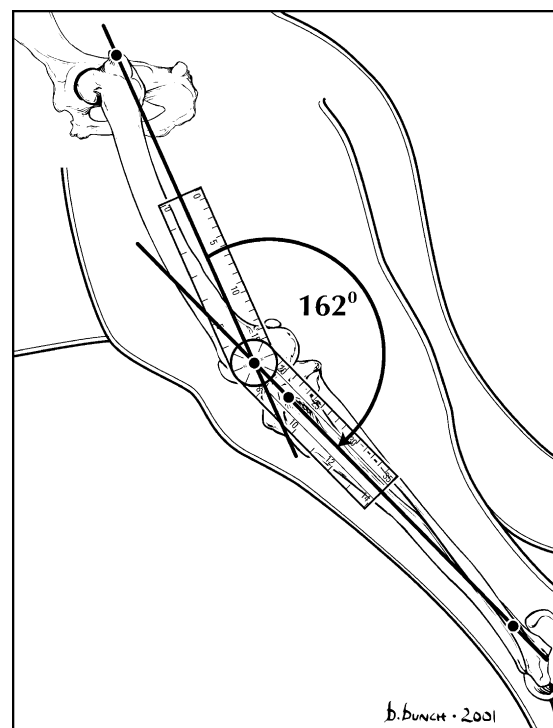
**Figure 11** Tarsal flexion.



**Figure 12** Tarsal extension.

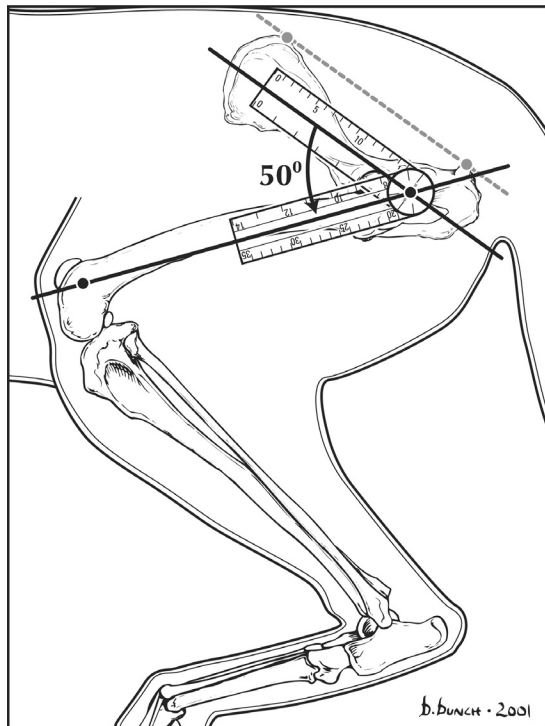


**Figure 13** Stifle joint flexion.

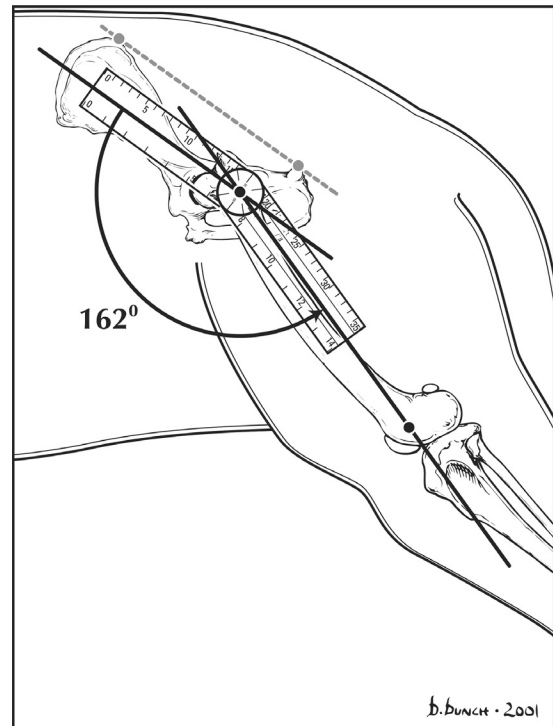


**Figure 14** Stifle joint extension.





**Figure 15** Hip joint flexion.



**Figure 16** Hip joint extension.

**Table 1** Range of Motion (degrees) of Various Appendicular Joints Measured by Goniometry in 16 Healthy Labrador Retrievers

| Joint     | Position | Mean | SD      | 95% CI of the Mean | Median |
|-----------|----------|------|---------|--------------------|--------|
| Carpus    | Flexion  | 32   | 2       | 31-34              | 32     |
| Extension | 196      | 2    | 194-197 | 196                |        |
| Valgus    | 12       | 2    | 11-13   | 12                 |        |
| Varus     | 7        | 1    | 6-8     | 7                  |        |
| Elbow     | Flexion  | 36   | 2       | 34-38              | 36     |
| Extension | 165      | 2    | 164-167 | 166                |        |
| Shoulder  | Flexion  | 57   | 2       | 54-59              | 57     |
| Extension | 165      | 2    | 164-167 | 165                |        |
| Tarsus    | Flexion  | 39   | 2       | 37-40              | 38     |
| Extension | 164      | 2    | 162-165 | 165                |        |
| Stifle    | Flexion  | 42   | 2       | 40-43              | 41     |
| Extension | 162      | 3    | 160-164 | 162                |        |
| Hip       | Flexion  | 50   | 2       | 48-52              | 50     |
| Extension | 162      | 3    | 160-164 | 162                |        |

CI = Confidence interval.



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